Reinforcement Learning from Verifiable Rewards

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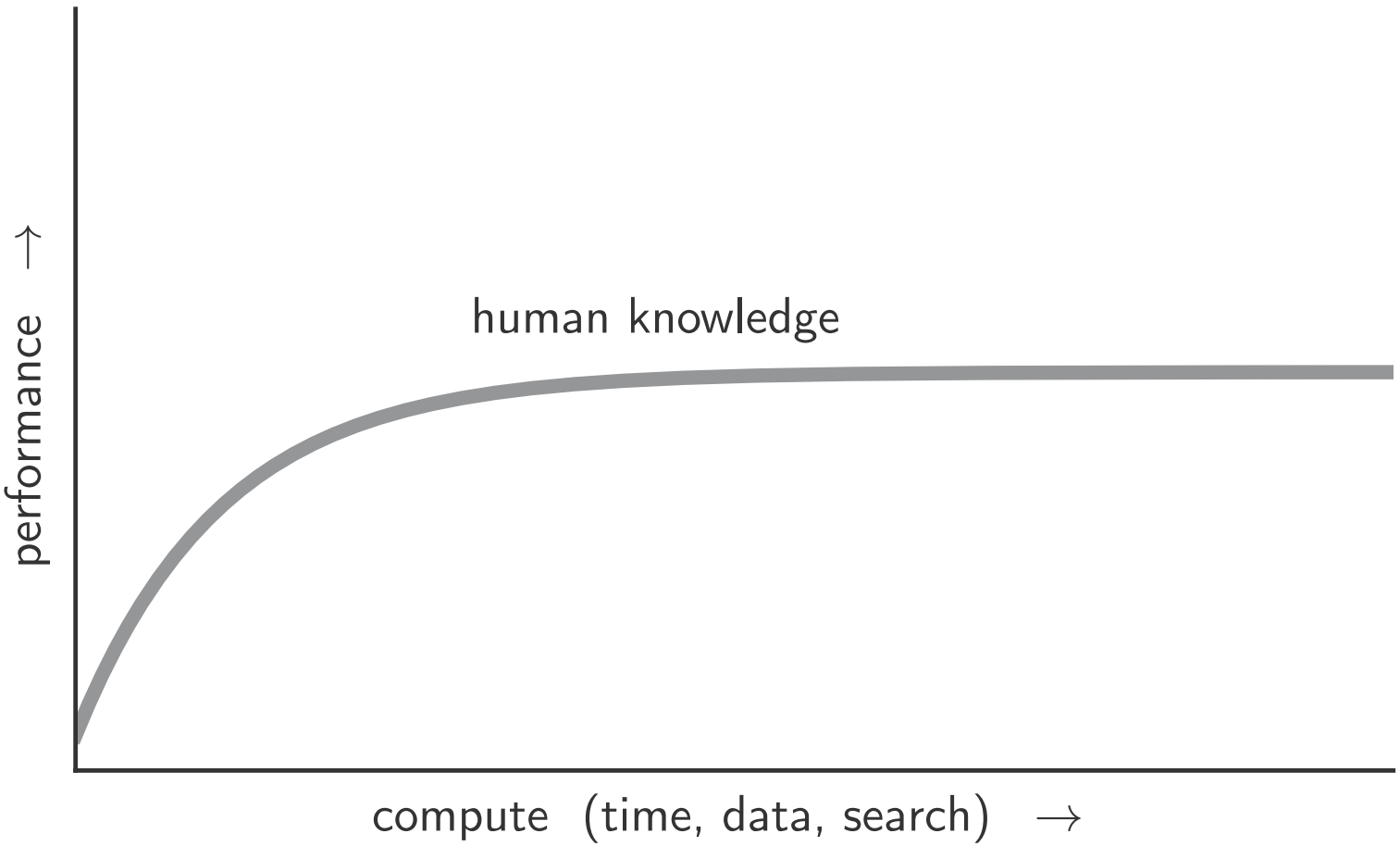
The Bitter Lesson

"The biggest lesson that can be read from 70 years of AI research is that general methods that leverage computation are ultimately the most effective, and by a large margin."

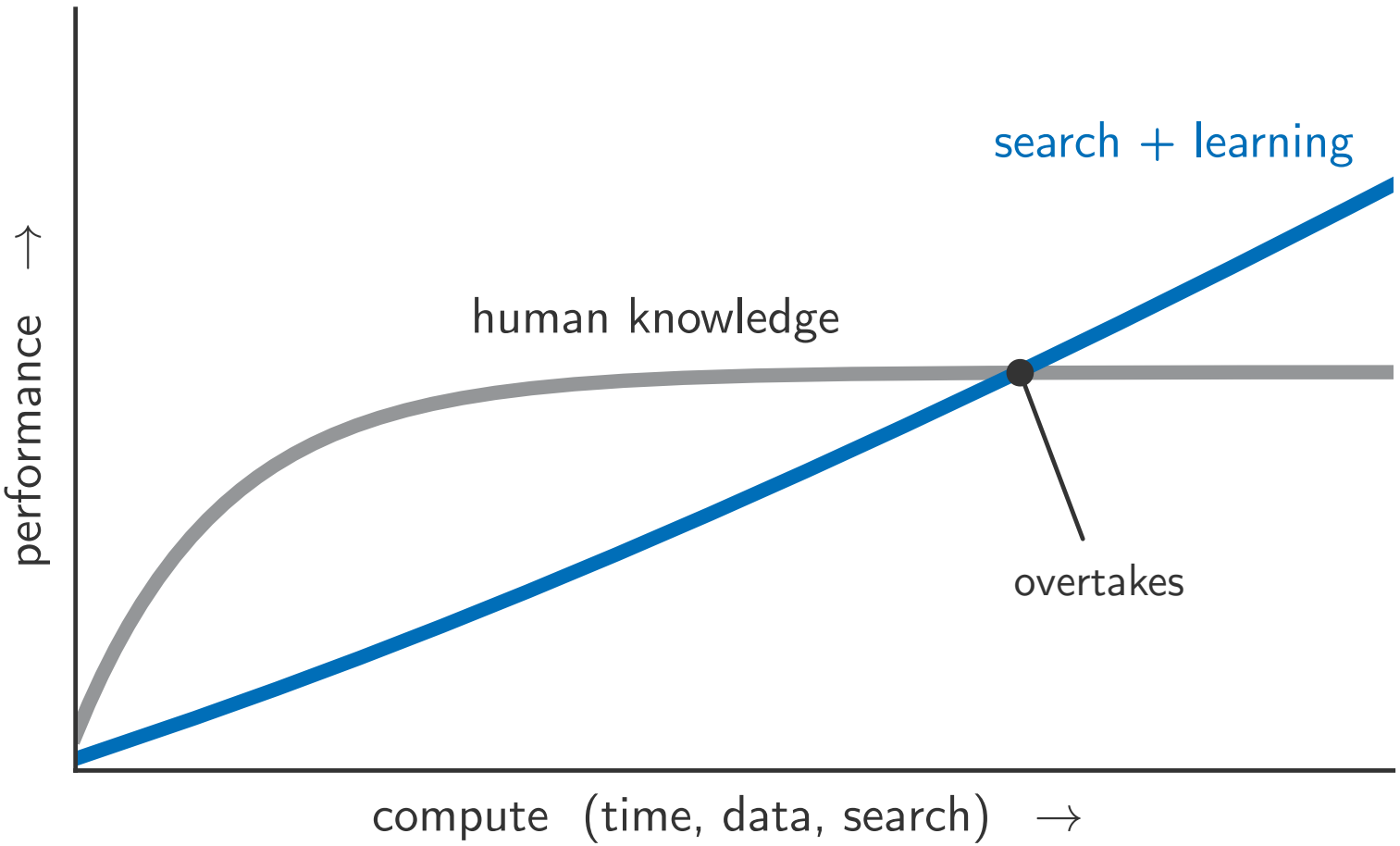
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What Scales?



What Scales?

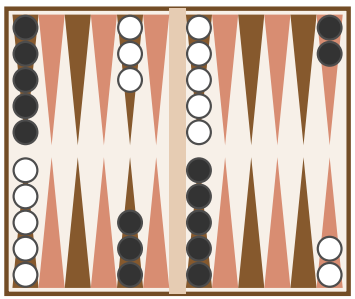
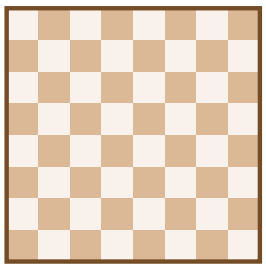
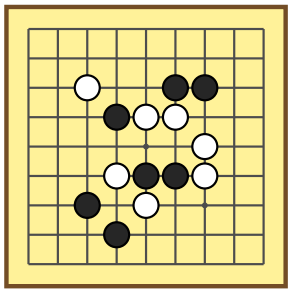
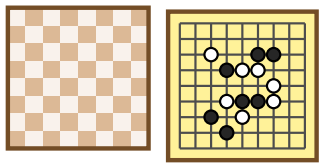

search + learning
human knowledge
overtakes
performance →
compute (time, data, search) →

Search and Learning

Human knowledge	**Search + Learning**
Contains what we discovered	**Discovers** from experience

Learning from self experience is the entire thesis of RL.

A Short History

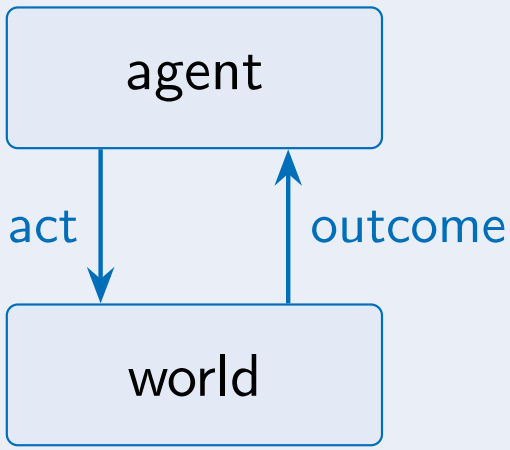
TD-Gammon | **Deep Blue** | **AlphaGo** | **AlphaZero**

1992	**1997**	**2016**	**2017**
self-play, learned value	search + hand-crafted eval	human games + self-play	self-play only

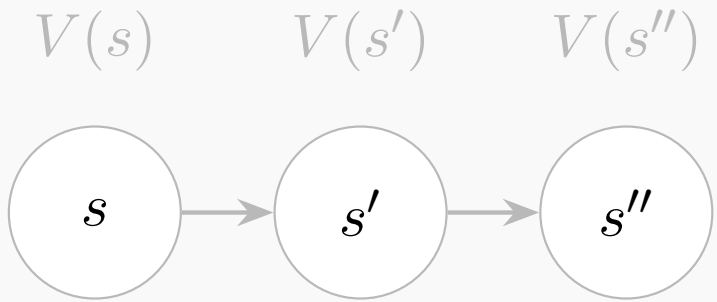
human knowledge → self-experience

Two Key Ideas

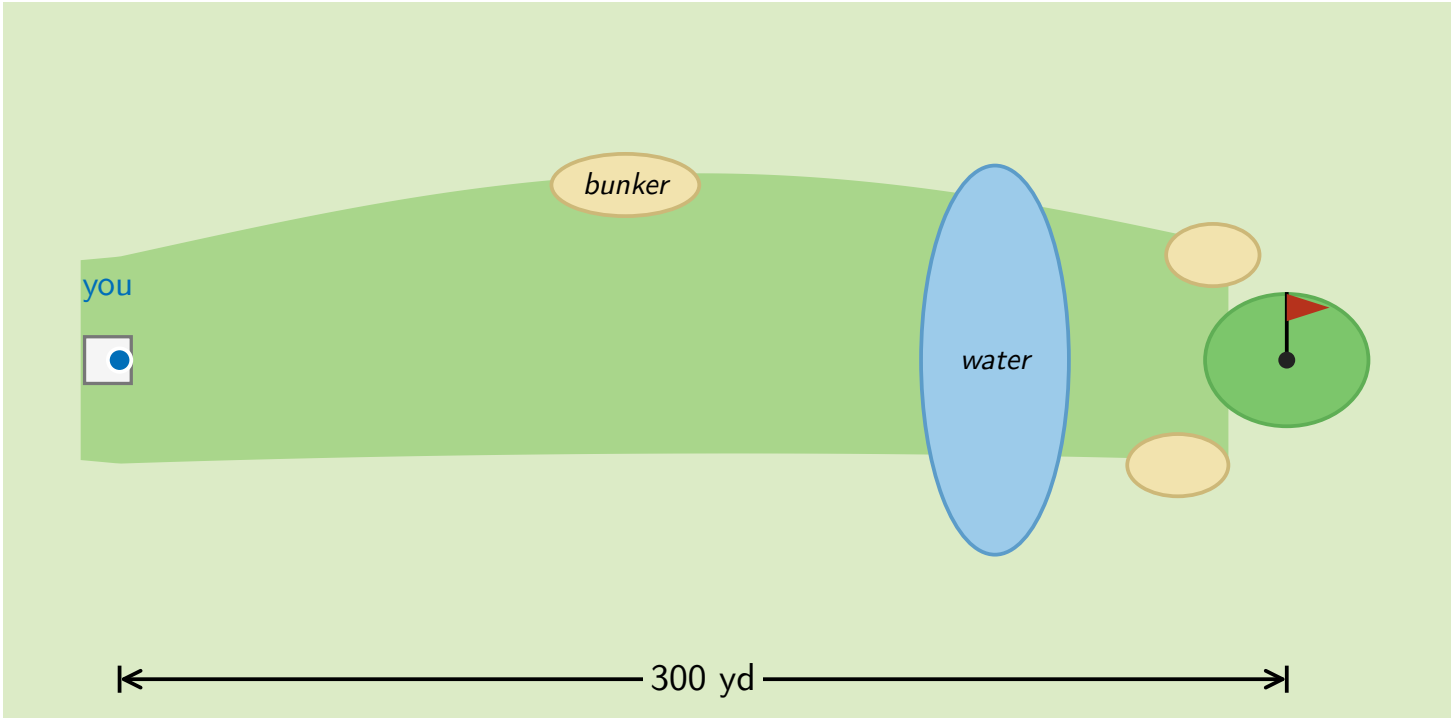
1. Learn from experience



2. Learn the value of behaviour

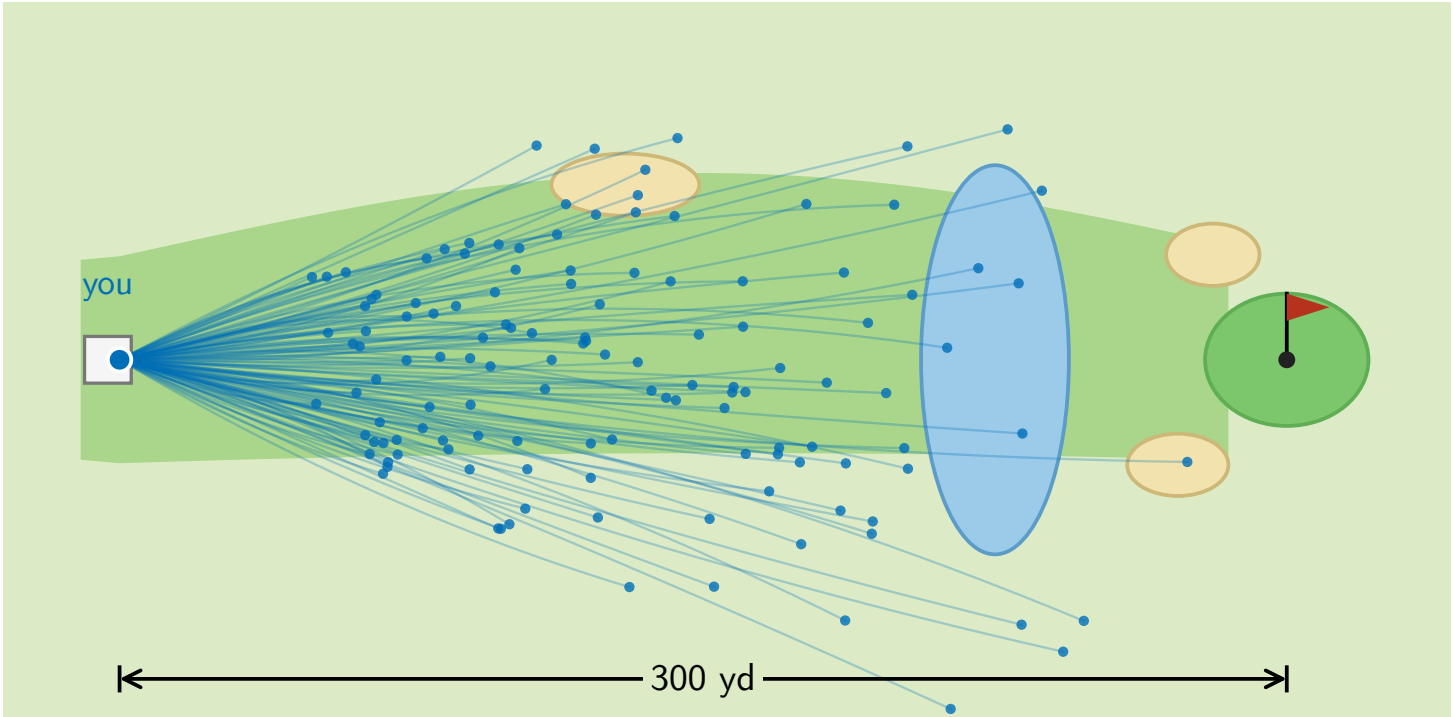


On the Tee

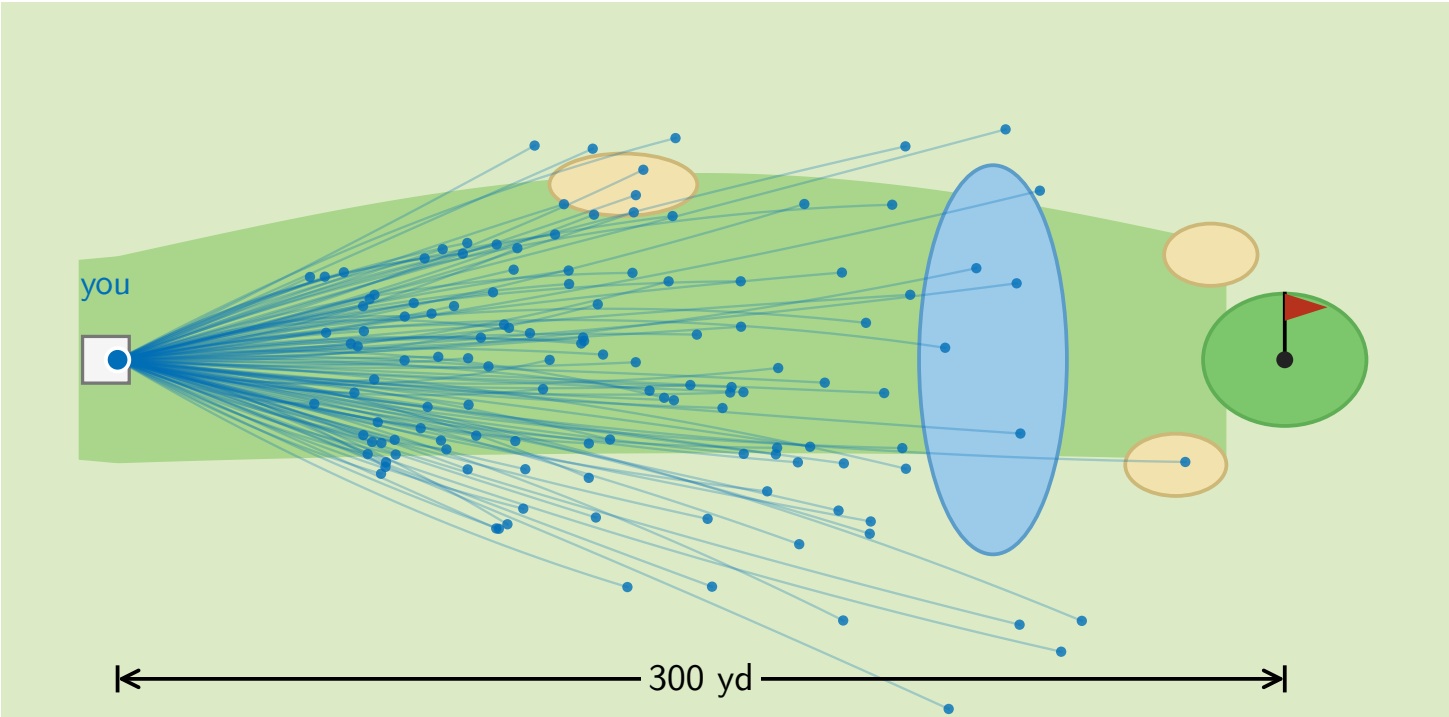


300 yards to the hole. What do you do?

So Many Choices

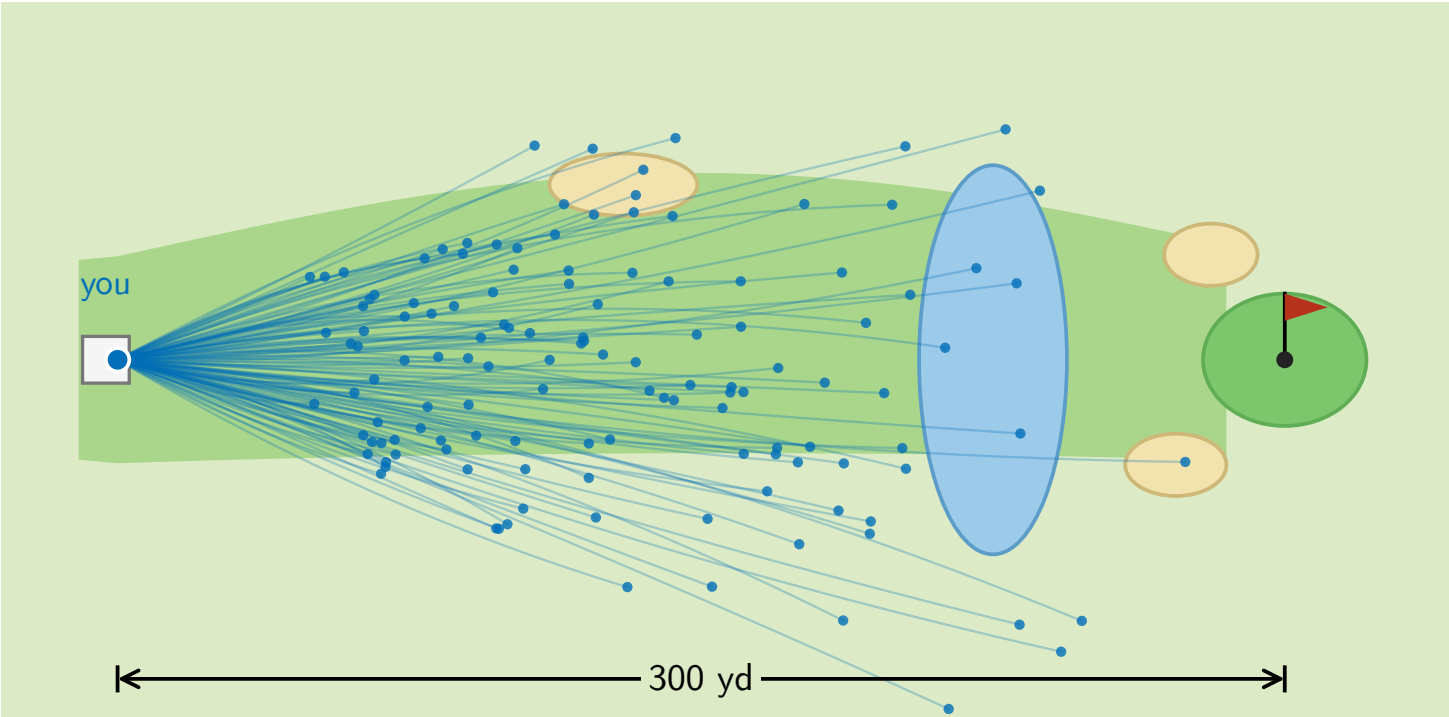


So Many Choices



14 clubs × 30 angles × 20 strengths

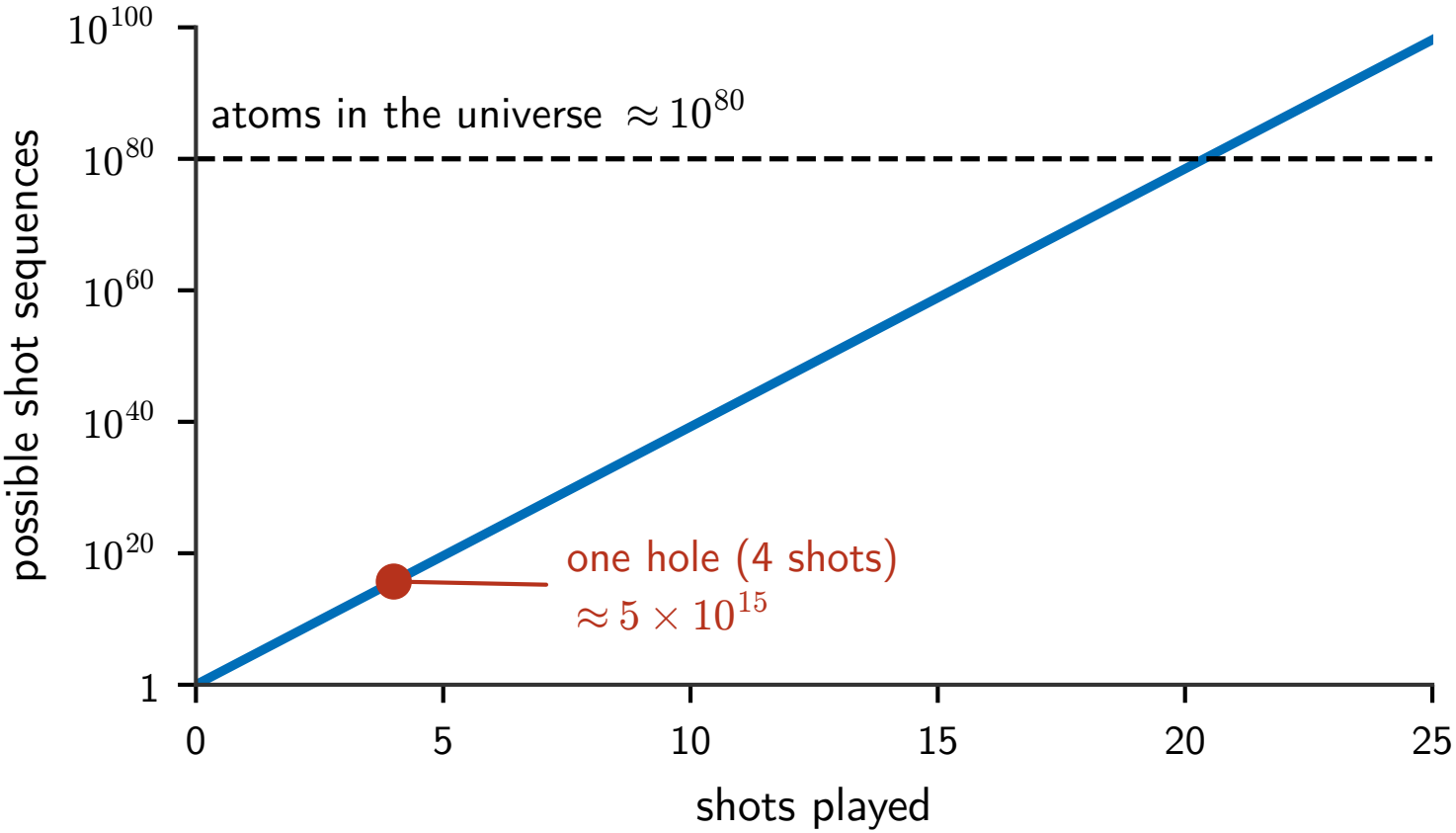
So Many Choices



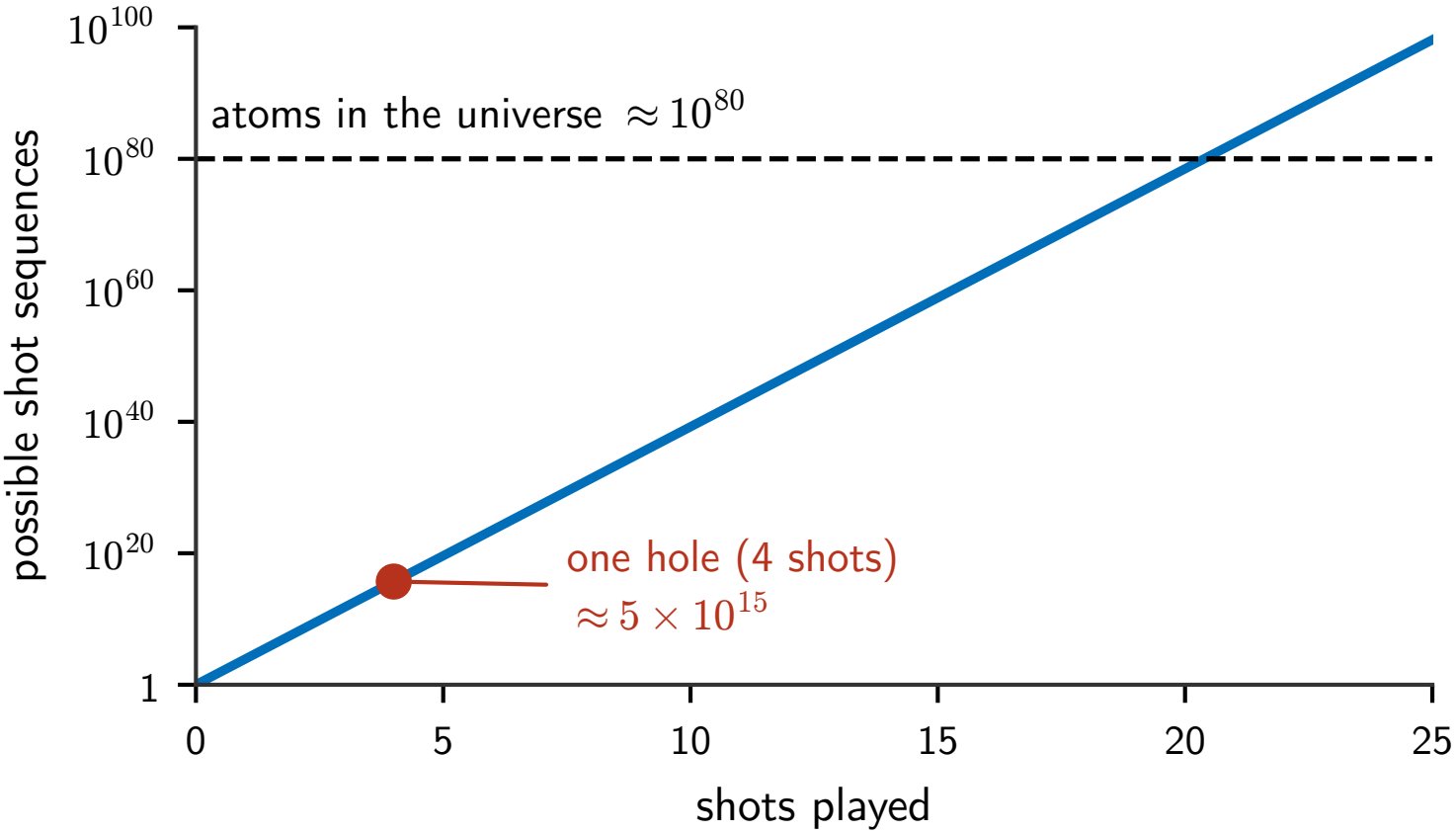
14 clubs × 30 angles × 20 strengths

$\approx 8{,}400$ actions per shot

Too Many Sequences

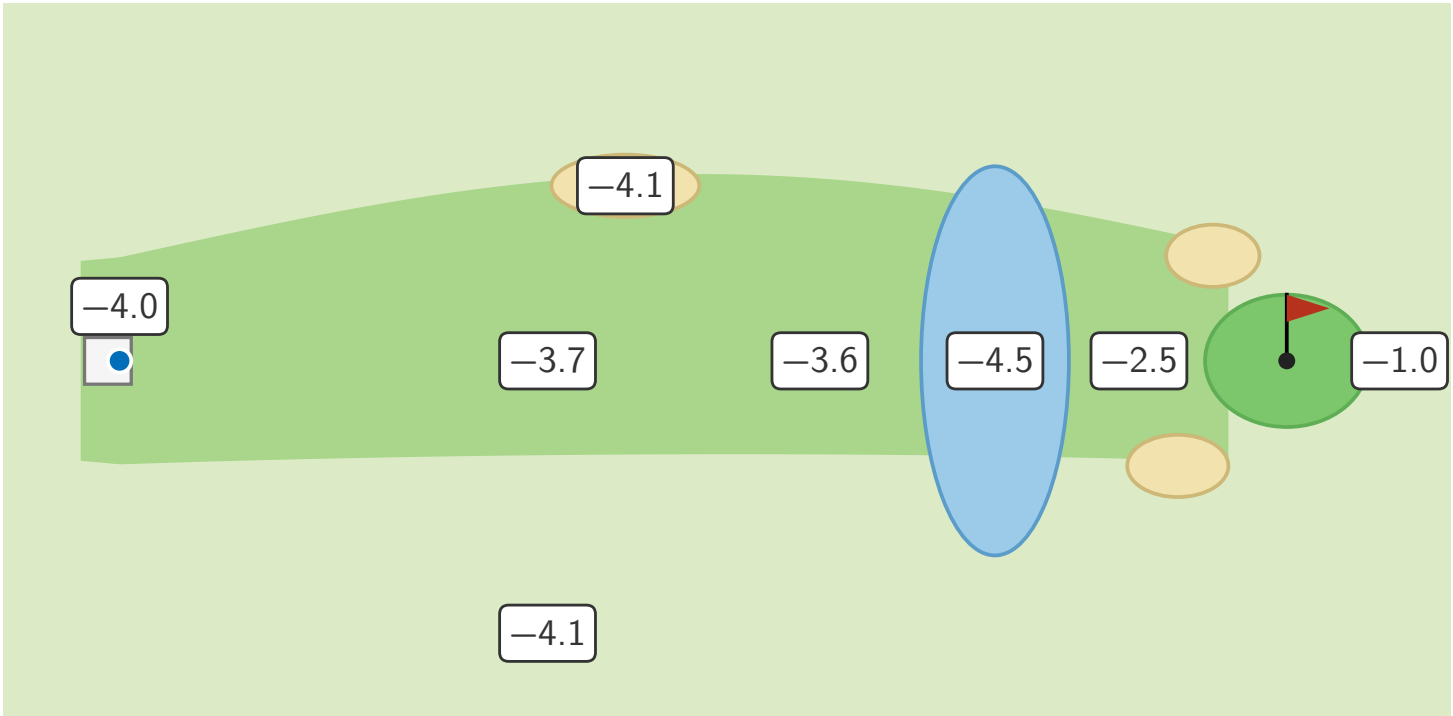


Too Many Sequences



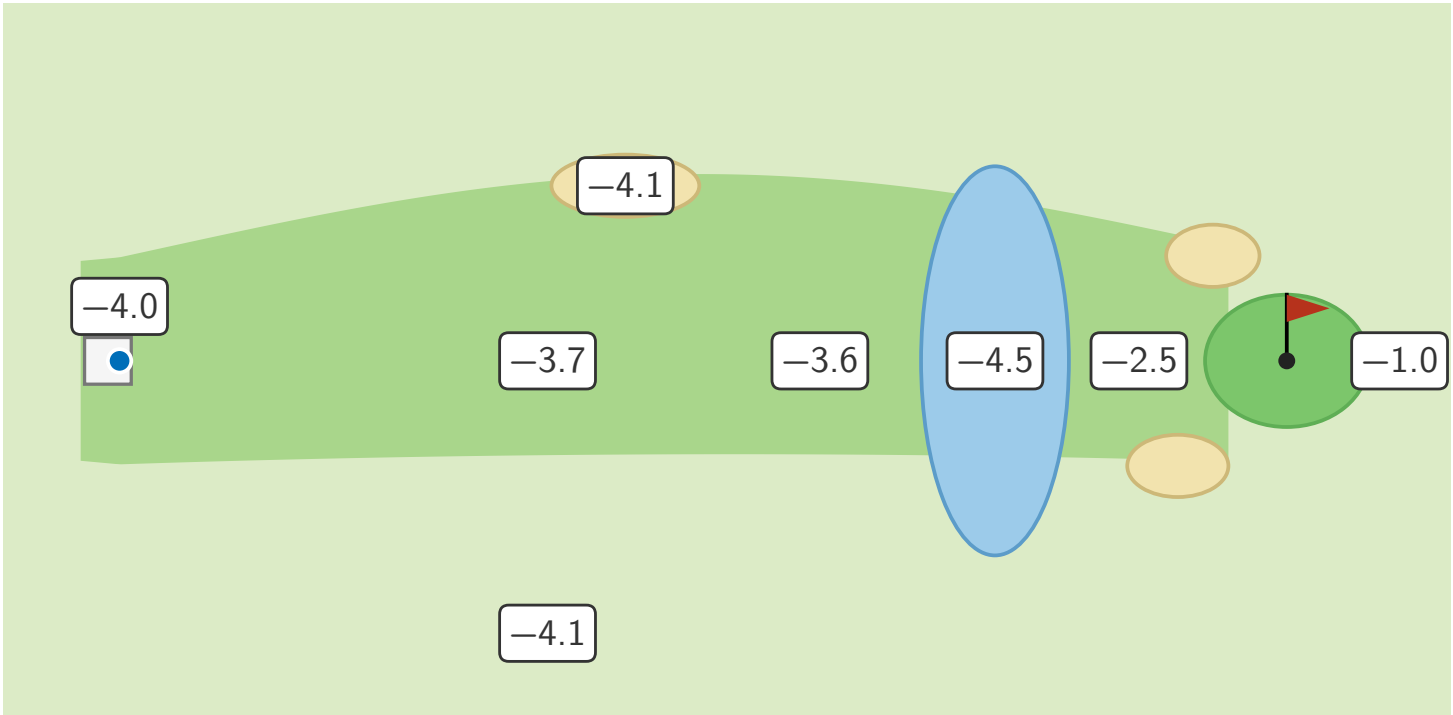
Learning the policy directly is intractable

Value of a State



$V(s)$ = how good is it to be *here*?

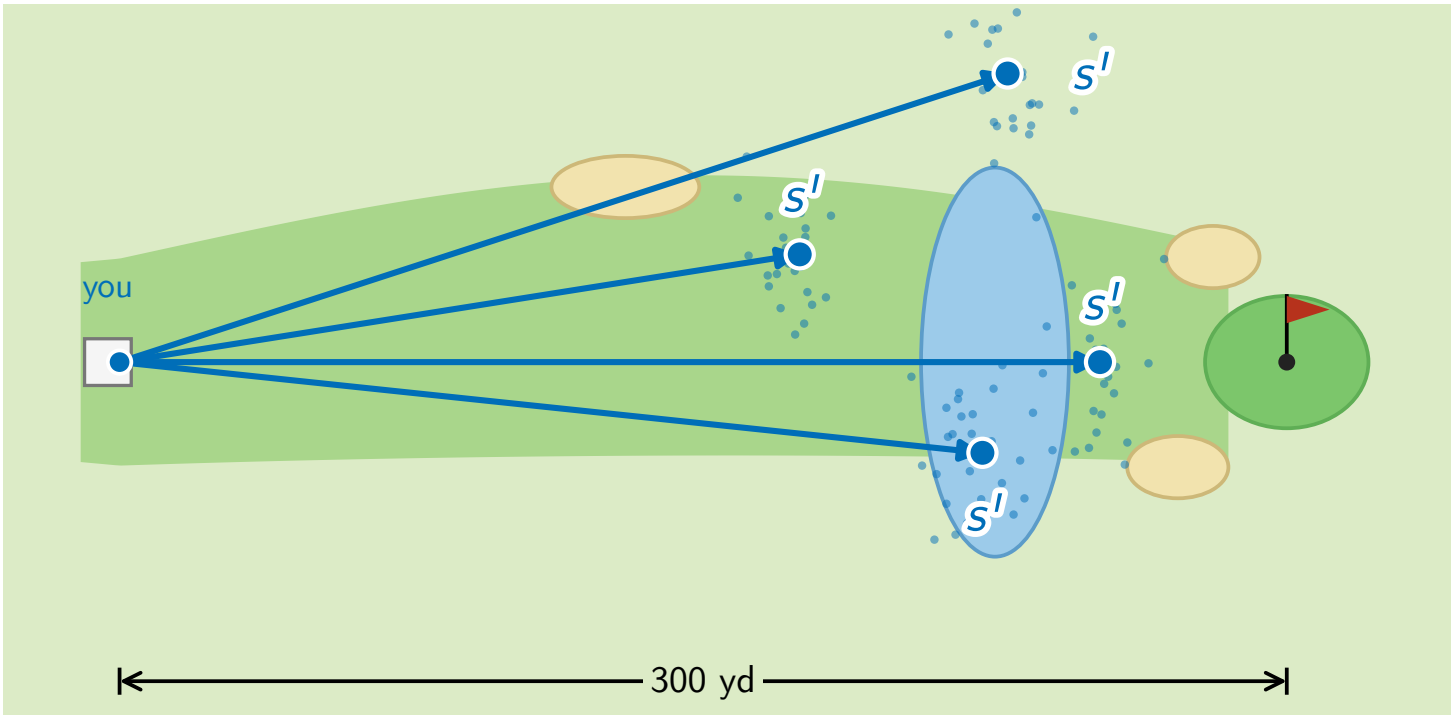
Value of a State



$V(s)$ = how good is it to be *here*?

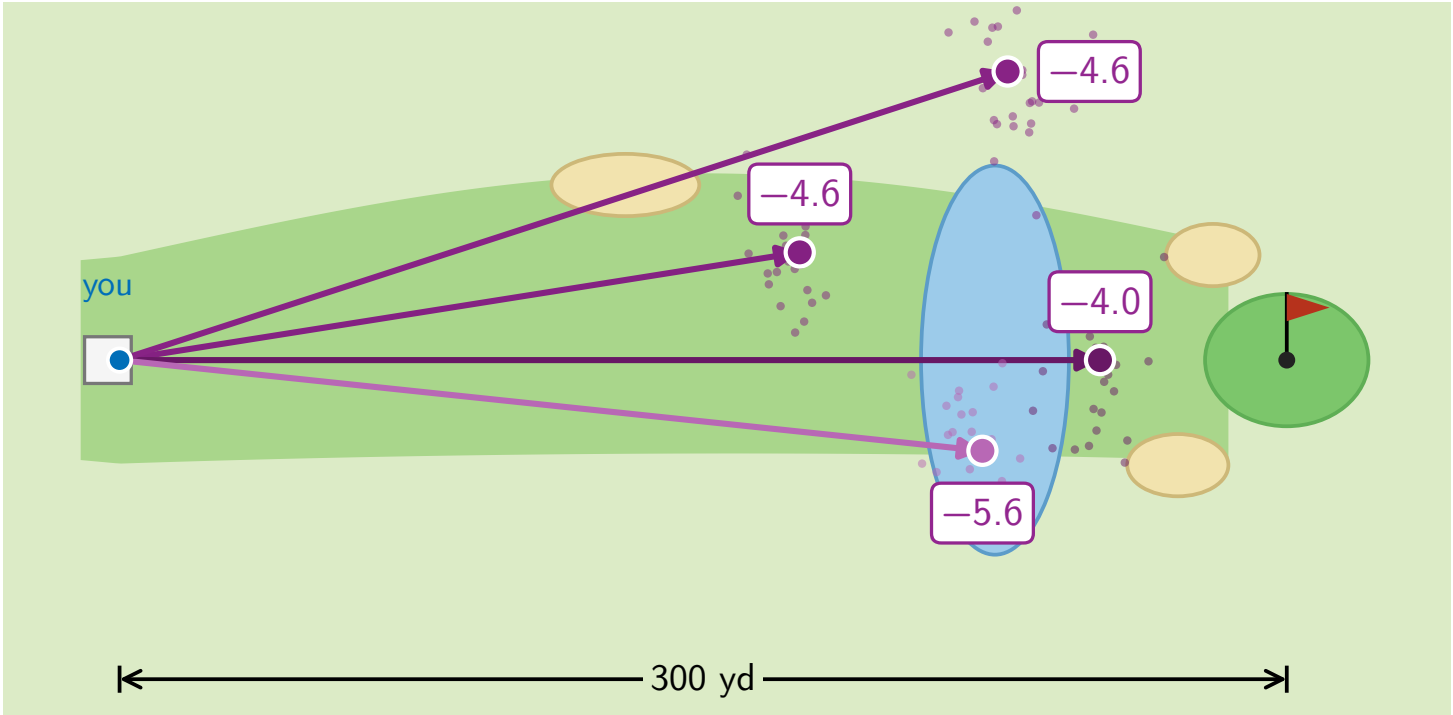
$V(s) \approx -$ expected strokes to finish the hole from s

Where Does It Land?



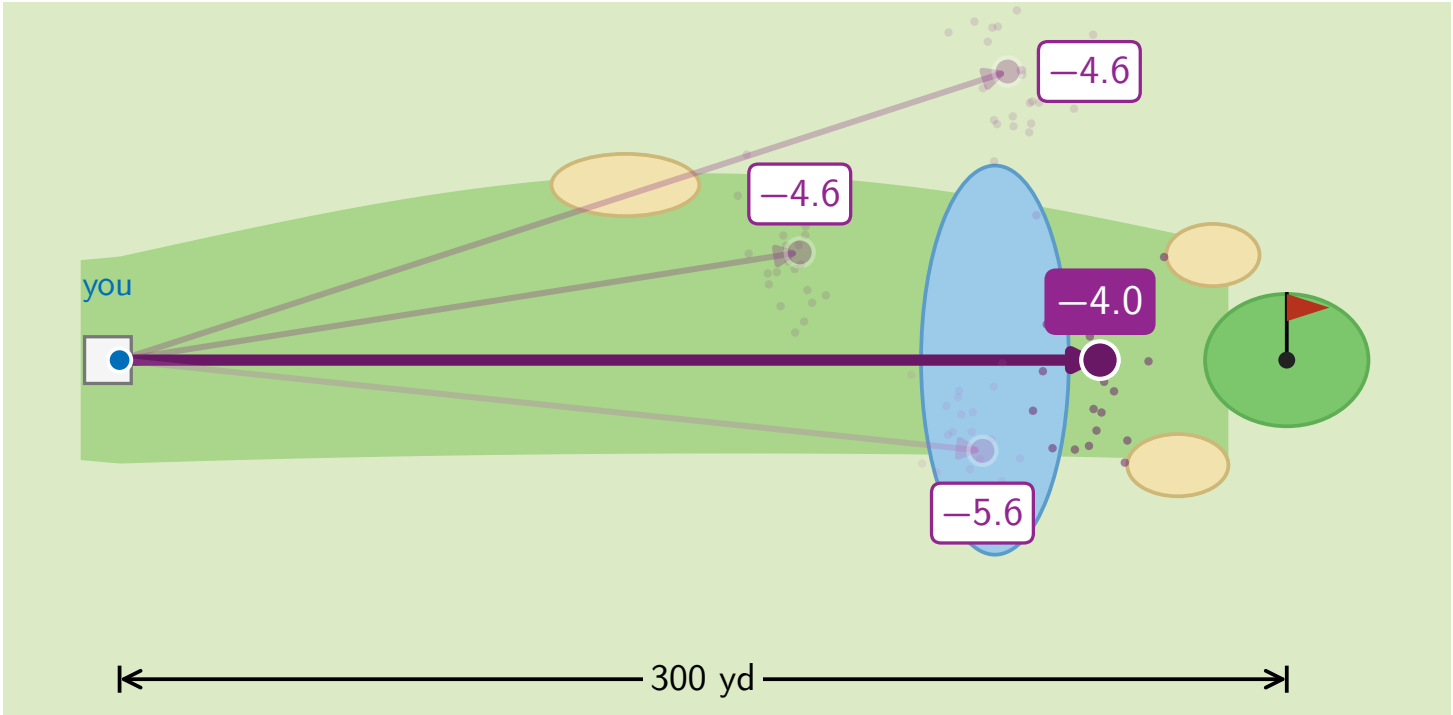
Each shot leads to a new state s'.

Where Does It Land?



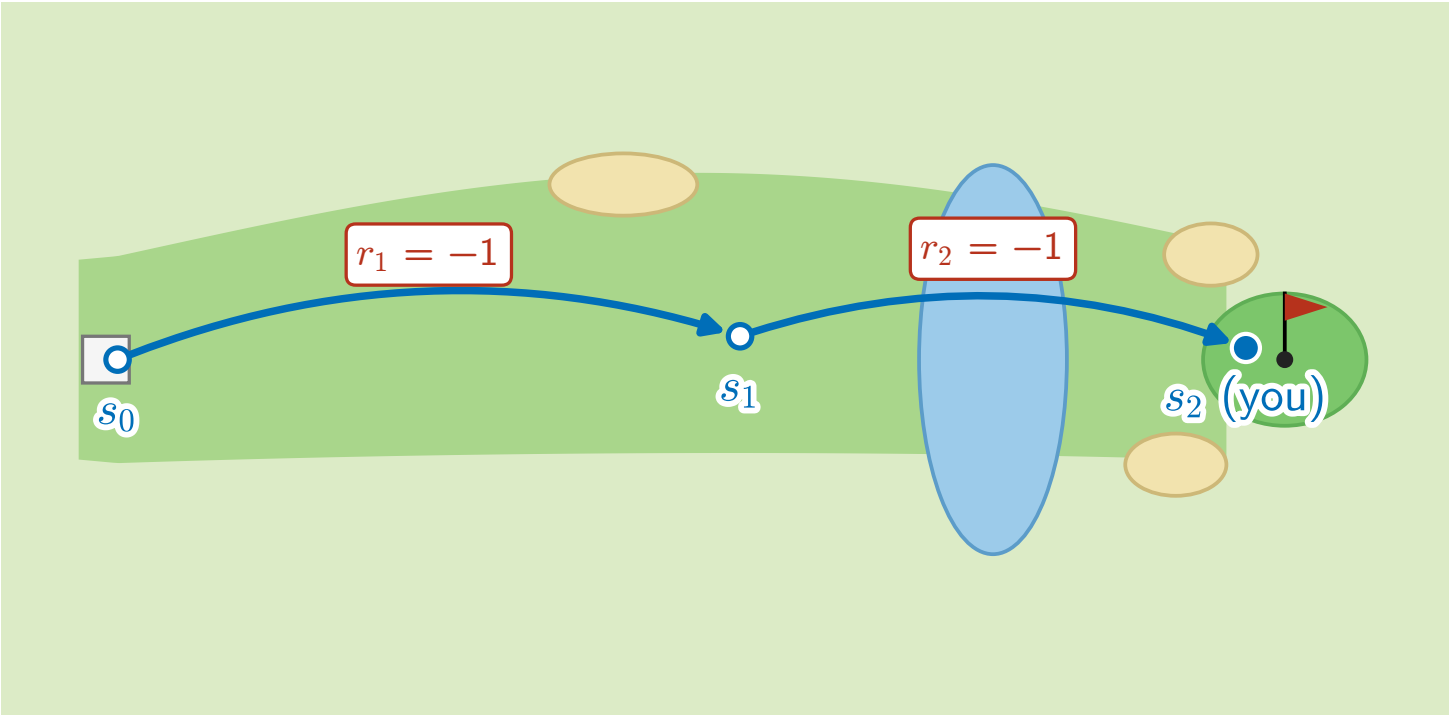
How good is each new state? $V(s')$

Where Does It Land?

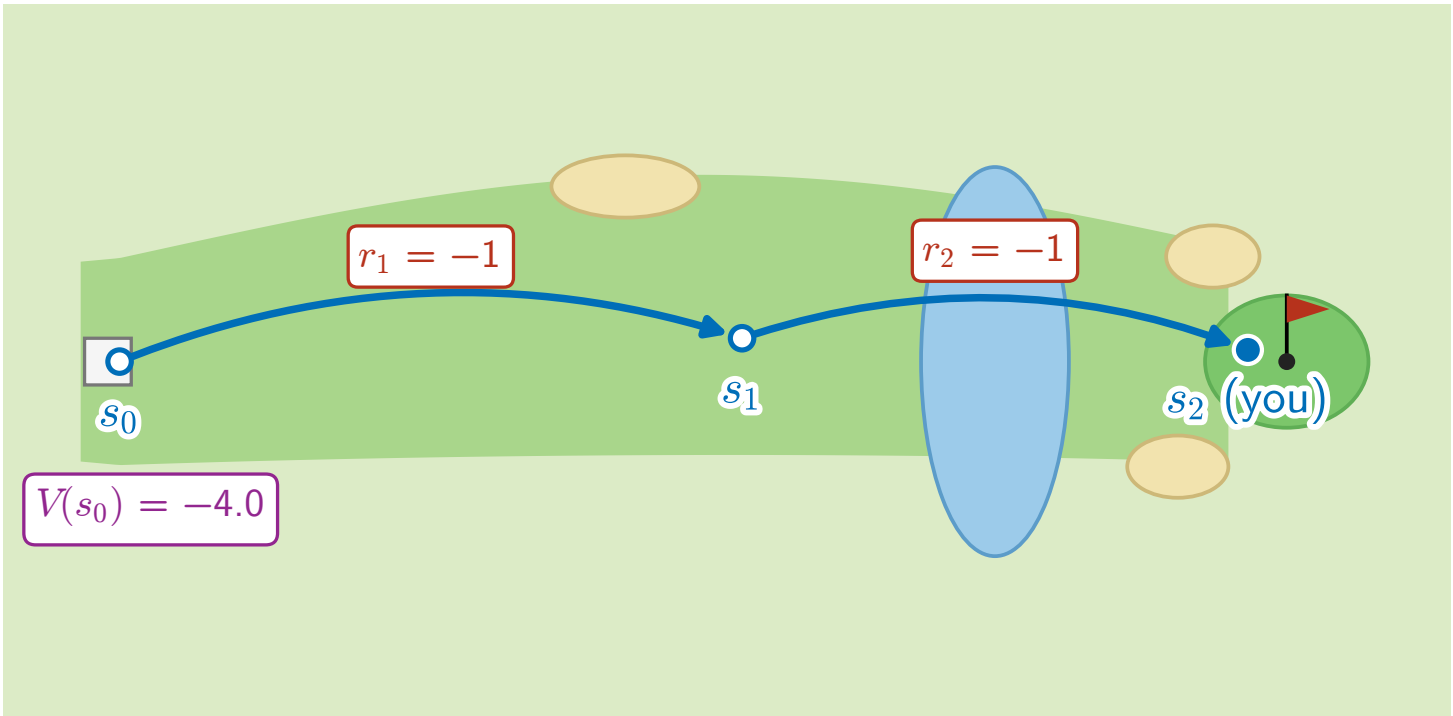


$$a^\star = \arg\max_a \left[r(s,a) + \gamma \sum_{s'} P(s' \mid s,a)\, V(s') \right]$$

What If I Can't Predict Where It Will Land?

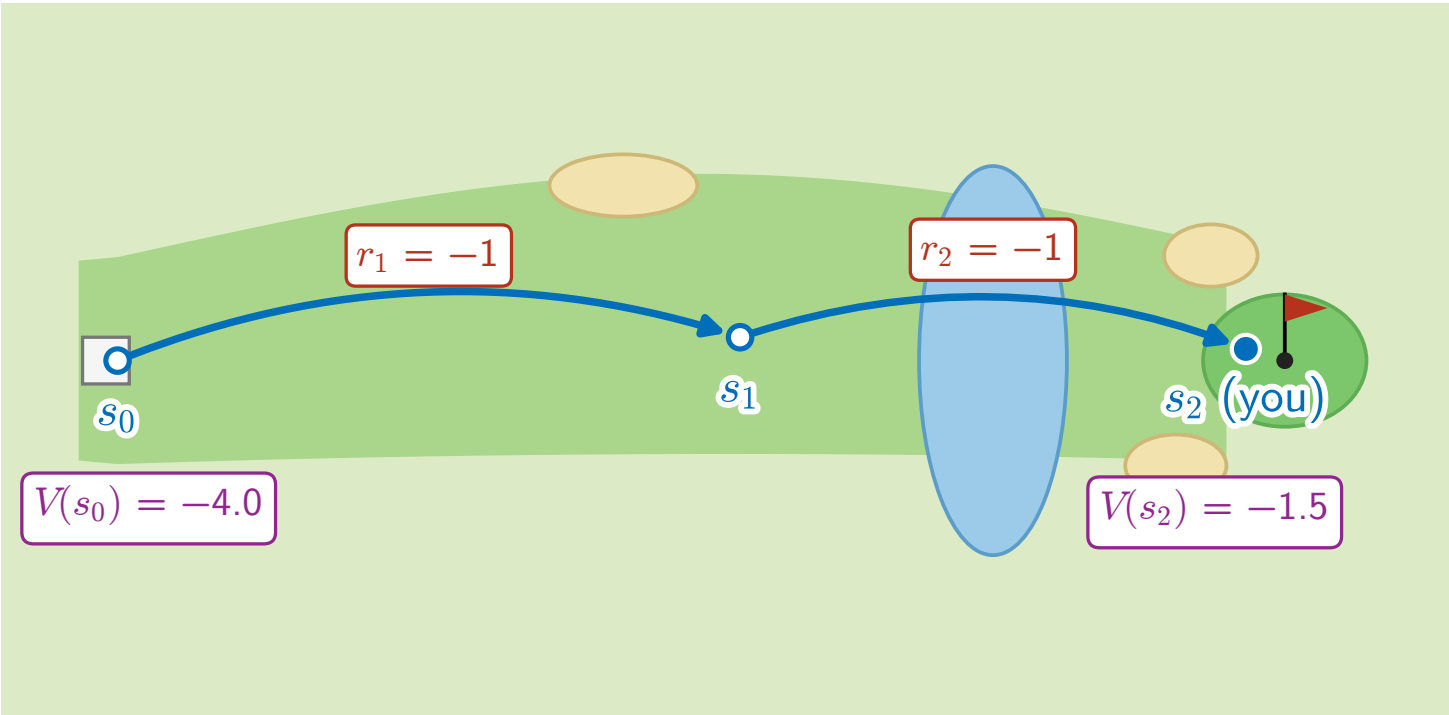


What If I Can't Predict Where It Will Land?



should have done: $V(s_0) = -4.0$

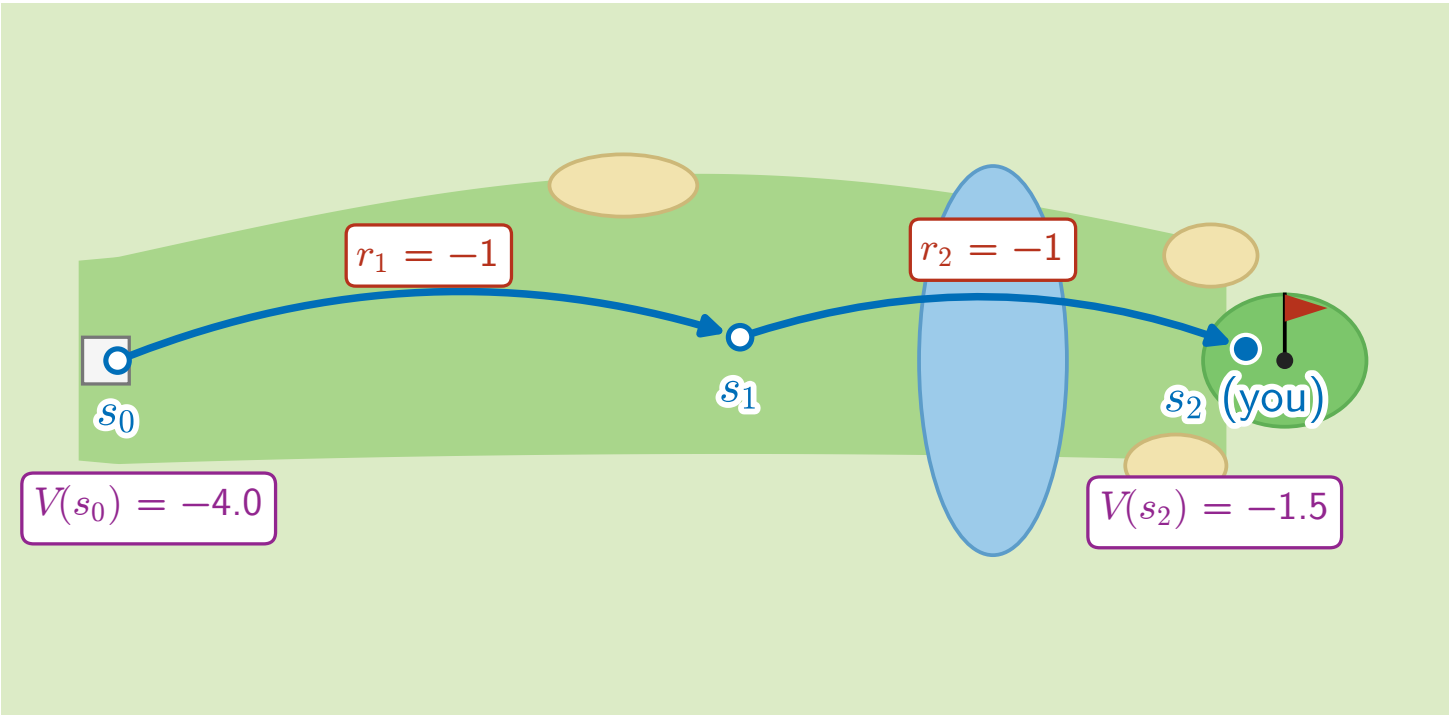
What If I Can't Predict Where It Will Land?



should have done: $V(s_0) = -4.0$

actually doing: $r_1 + r_2 + V(s_2) = -2 - 1.5 = -3.5$

What If I Can't Predict Where It Will Land?

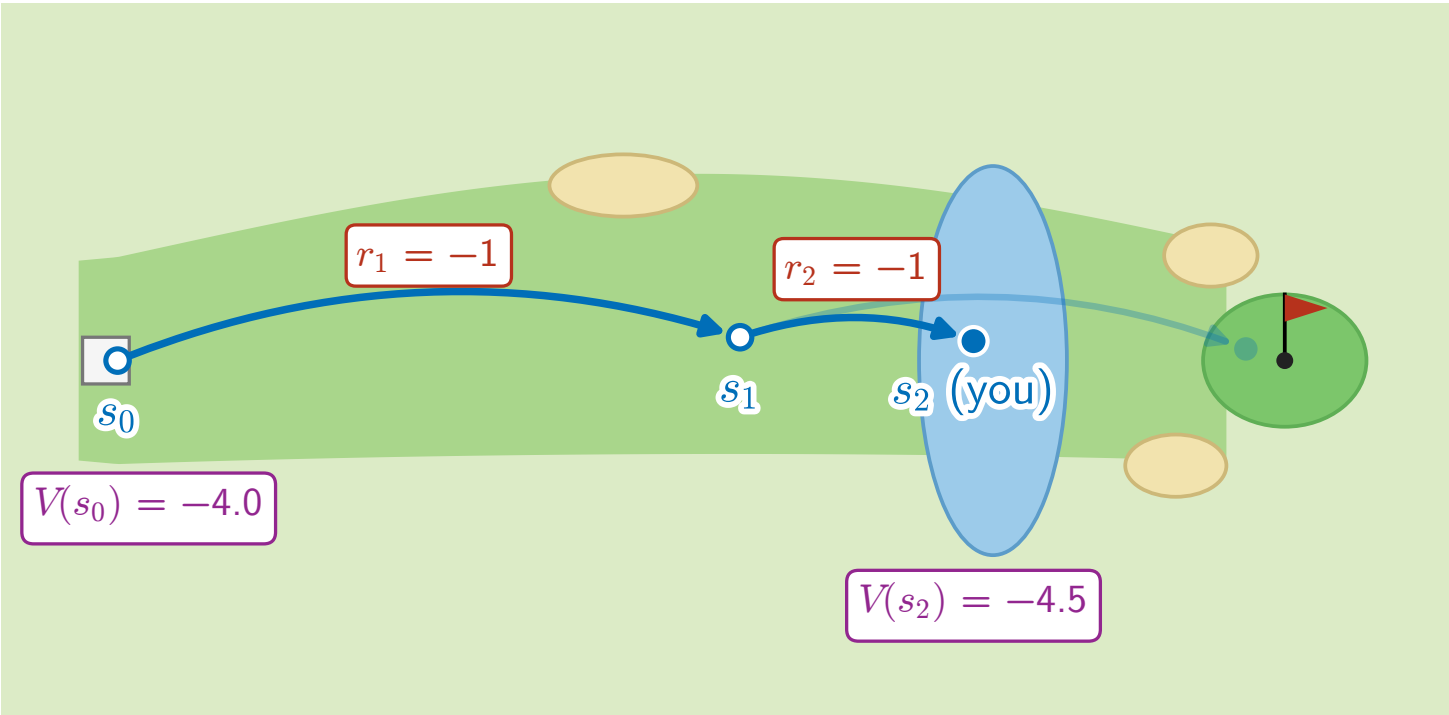


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actually doing: $r_1 + r_2 + V(s_2) = -2 - 1.5 = -3.5$

difference: $-3.5 - (-4.0) = +0.5$, better than expected!

What If I Can't Predict Where It Will Land?

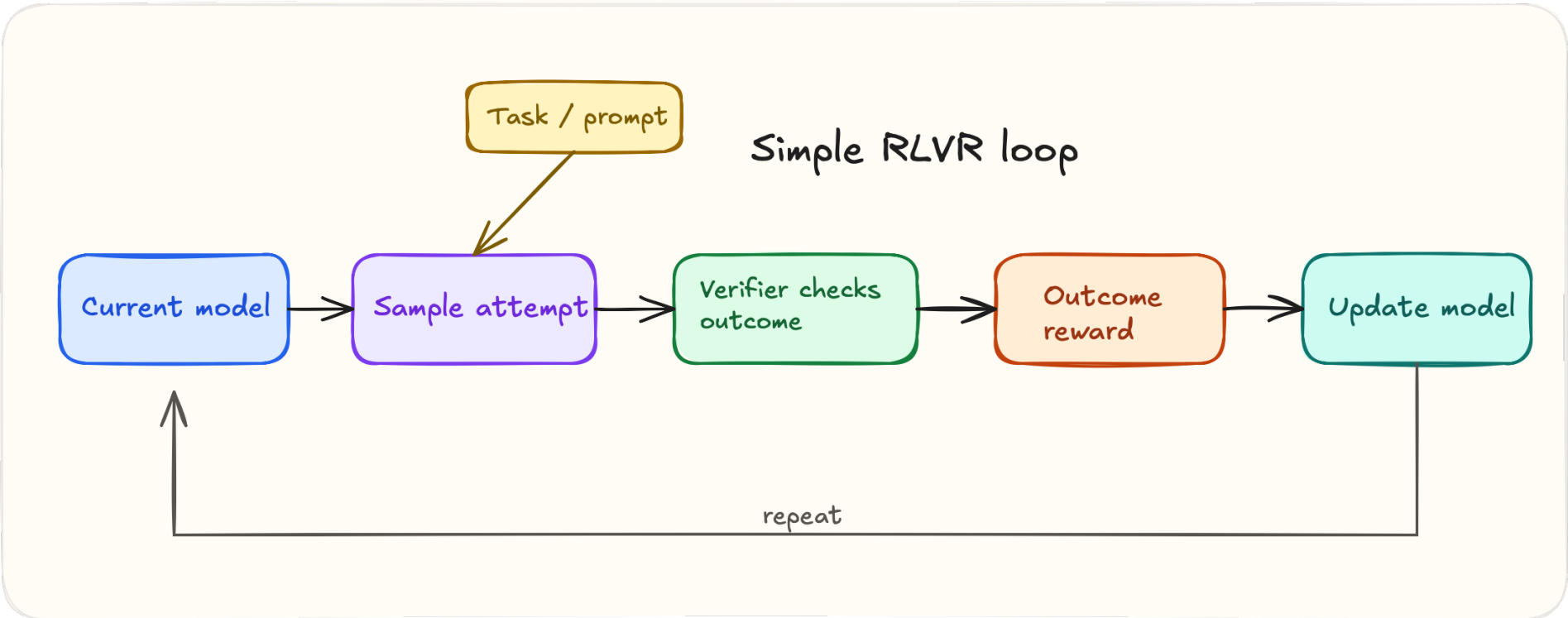


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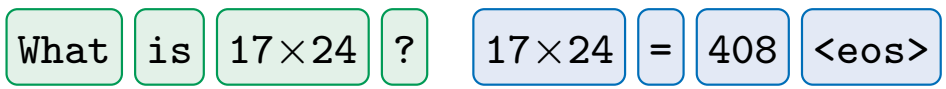
actually doing: $r_1 + r_2 + V(s_2) = -2 - 4.5 = -6.5$

difference: $-6.5 - (-4.0) = -2.5$, worse than expected!

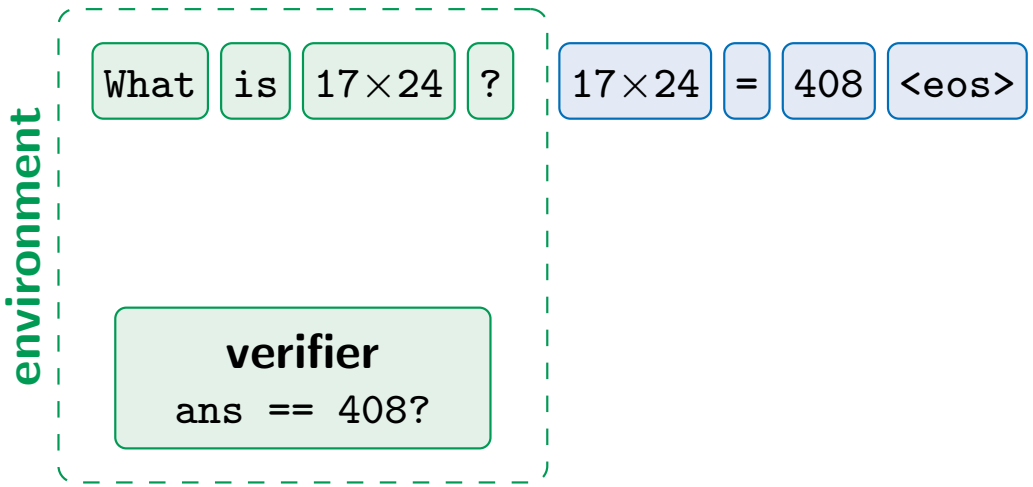
The RLVR Loop



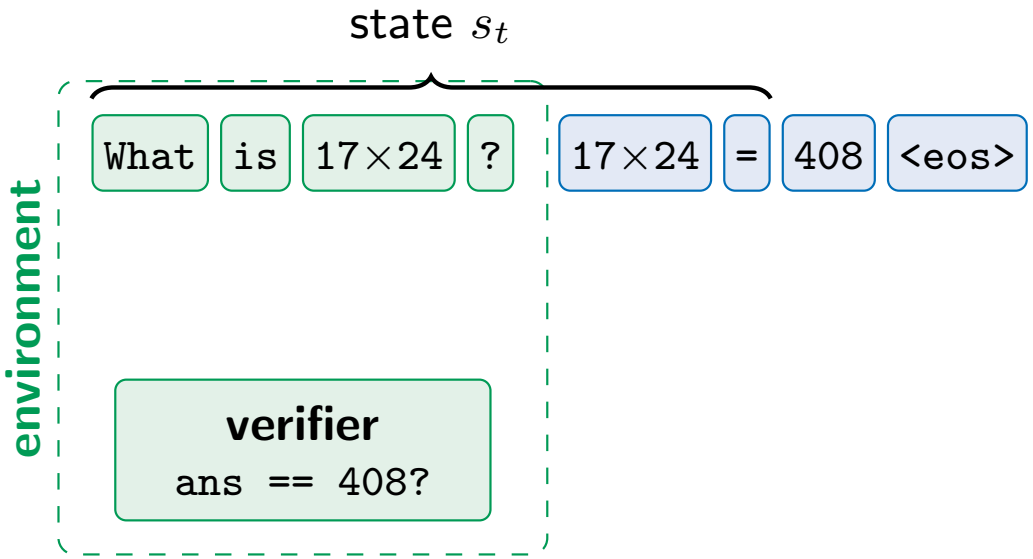
RL Concepts on an LLM



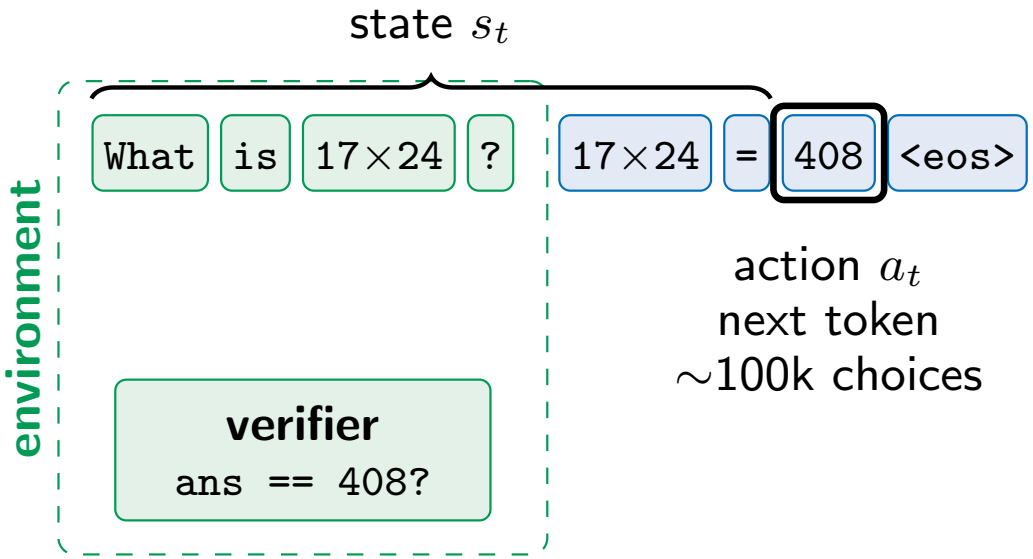
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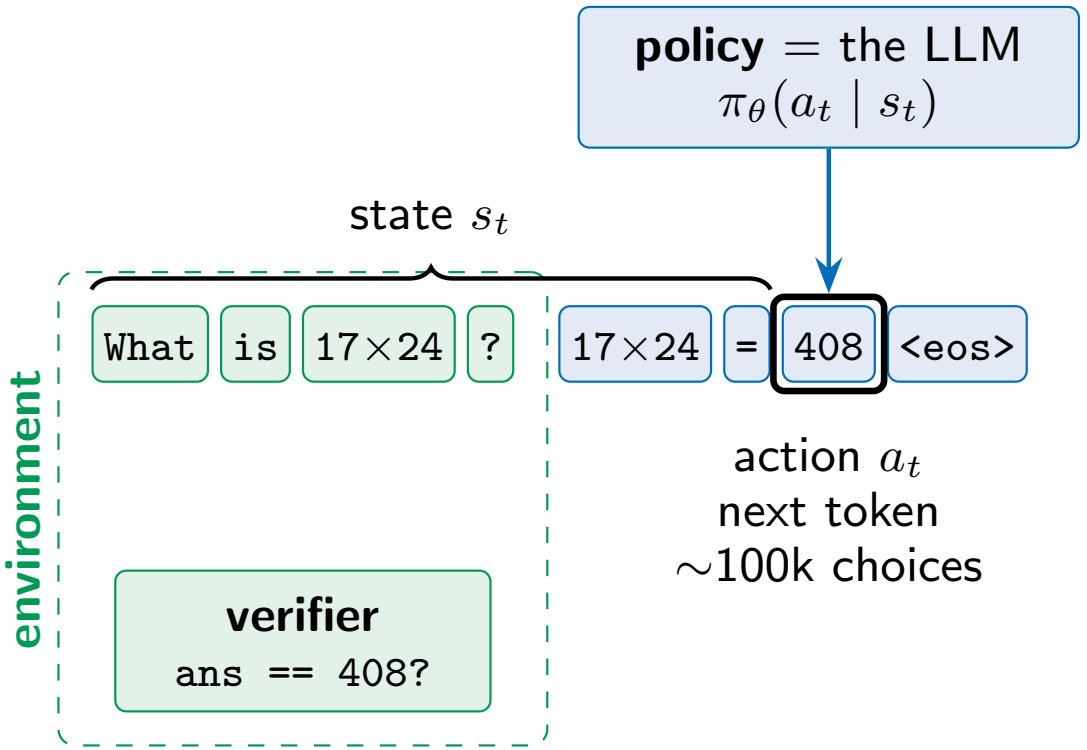
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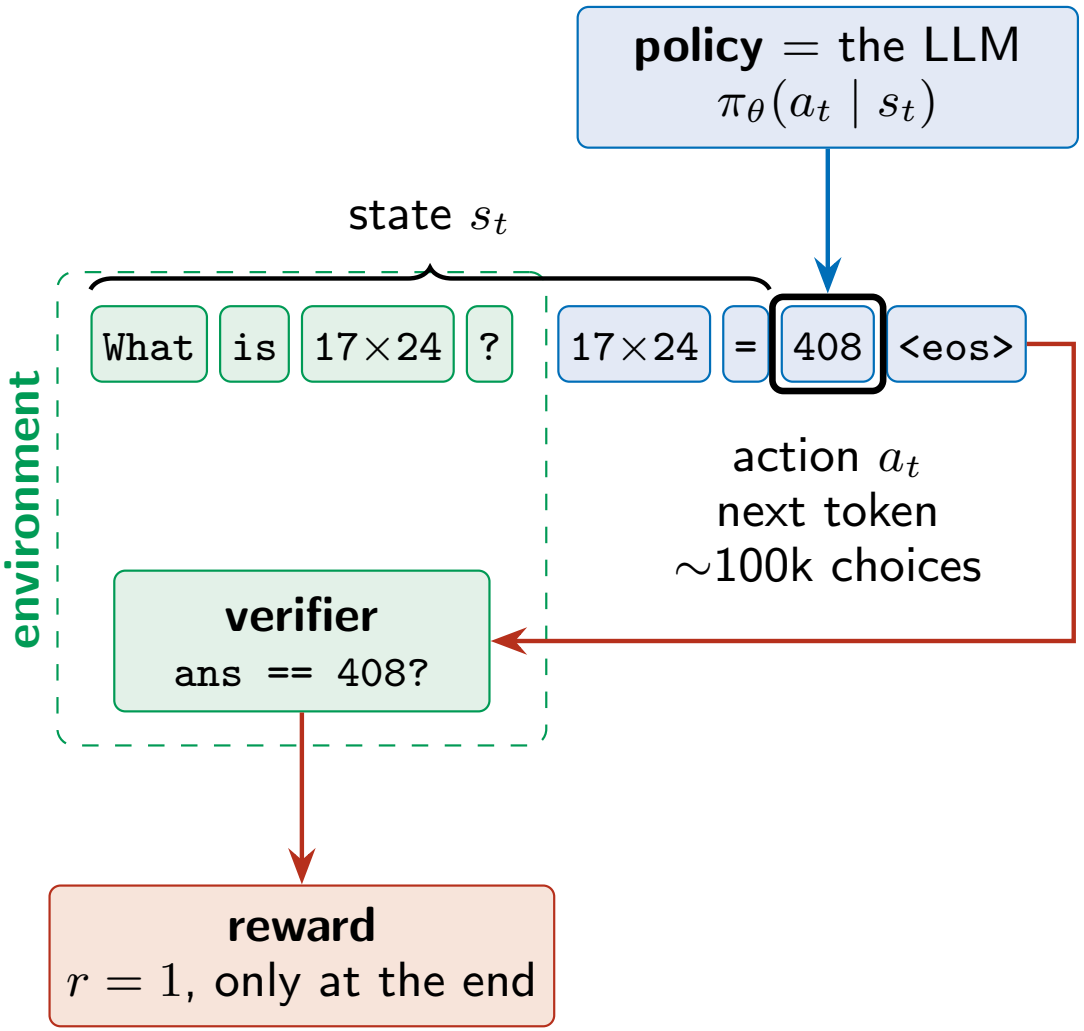
RL Concepts on an LLM



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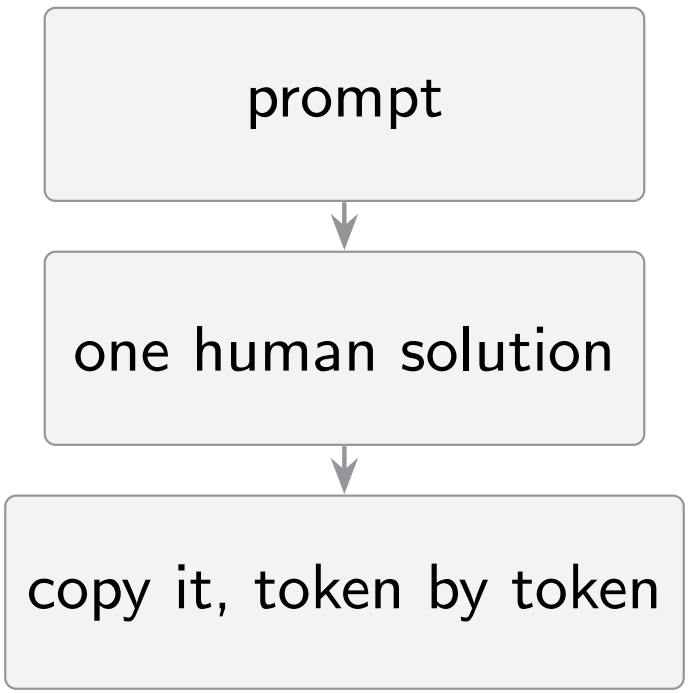


RL Concepts on an LLM

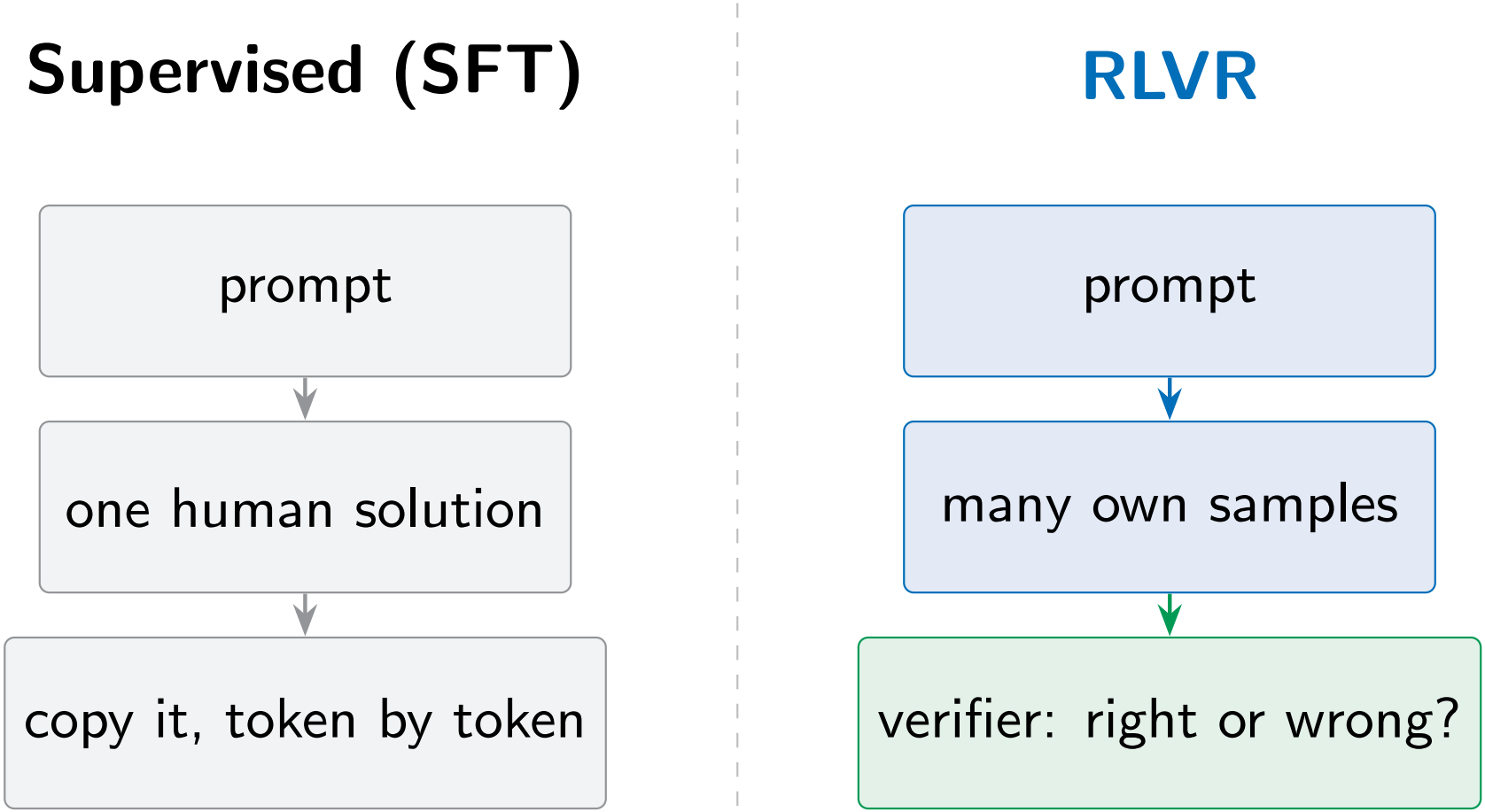


Why Not Just Imitate?

Supervised (SFT)



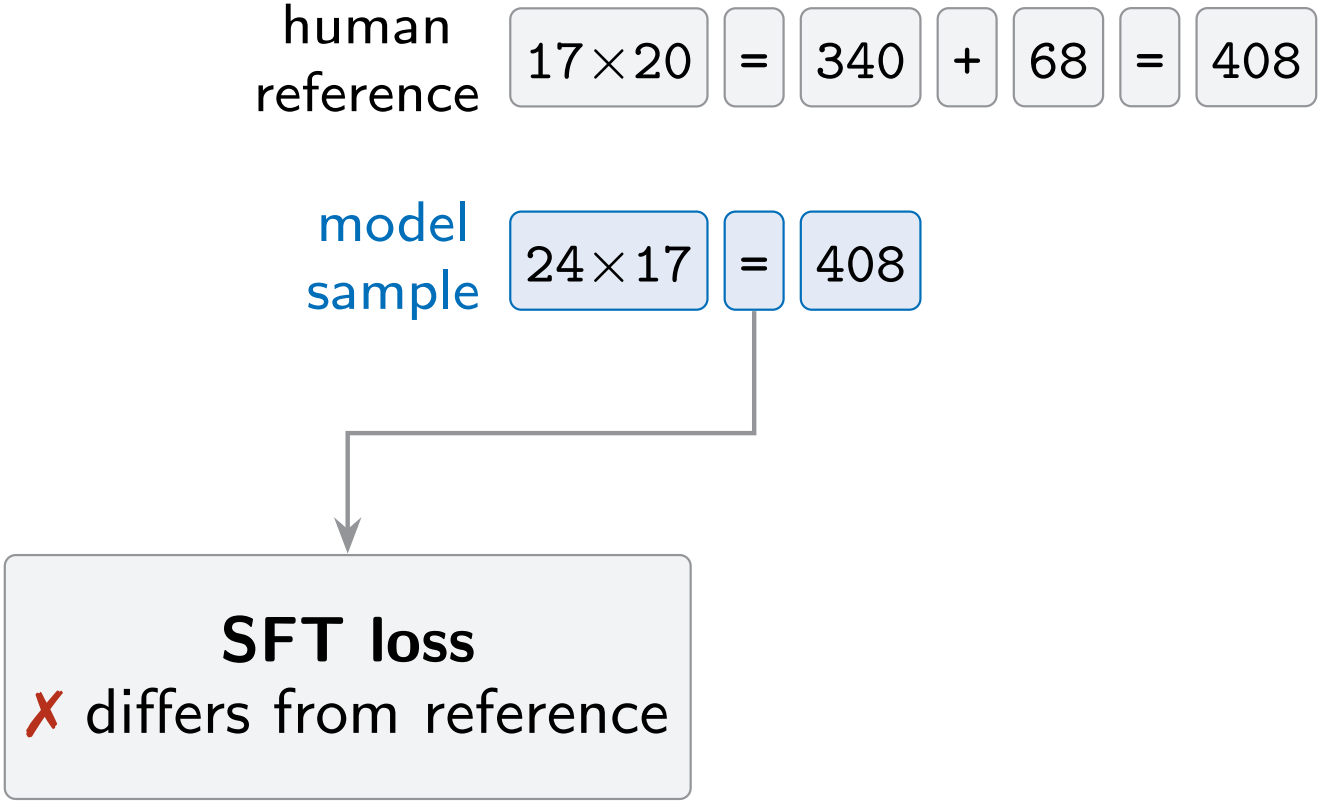
Why Not Just Imitate?



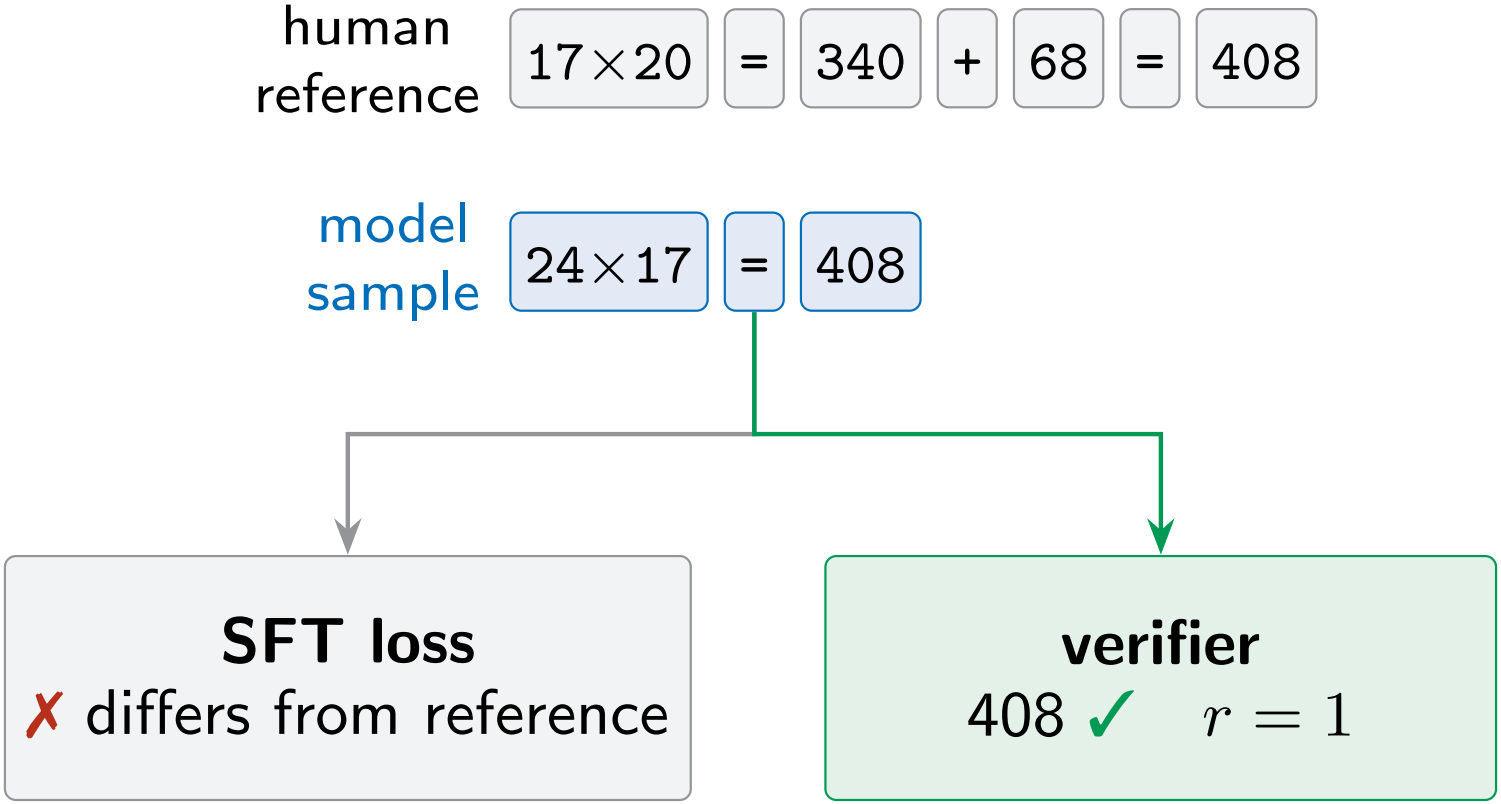
Right Answer, Different Words

human reference	17×20	=	340	+	68	=	408
model sample	24×17	=	408				

Right Answer, Different Words

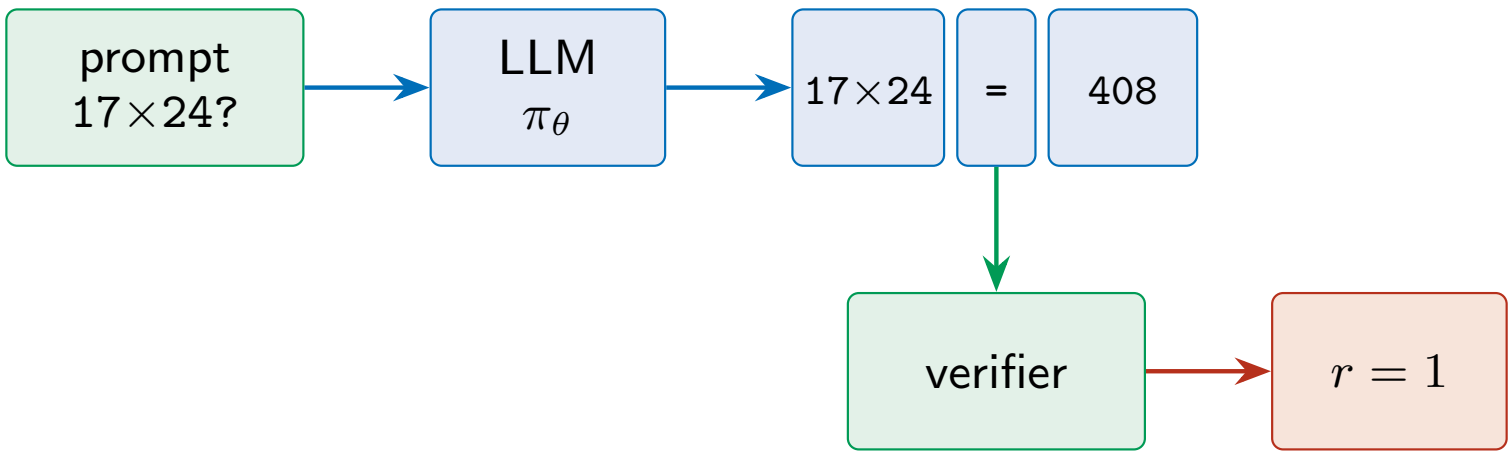


Right Answer, Different Words

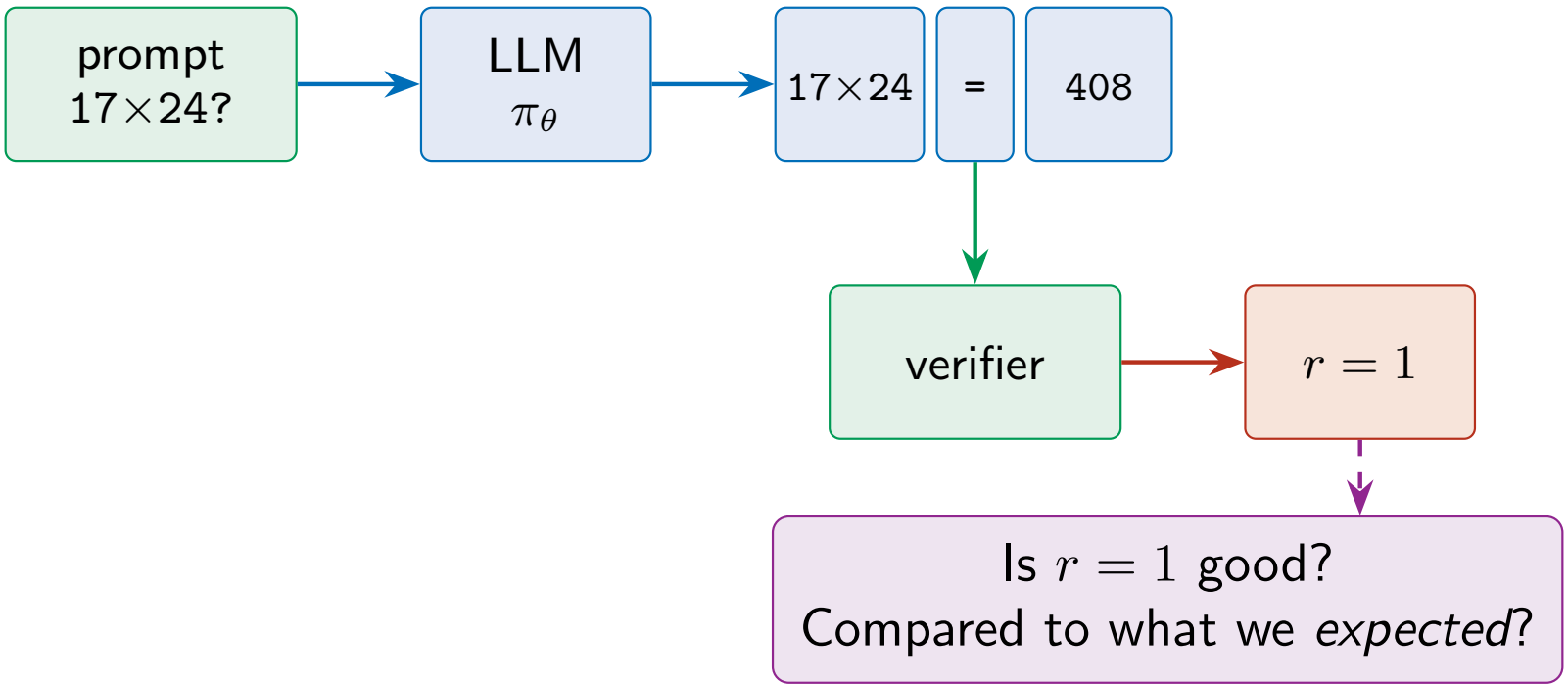


SFT rewards matching the teacher. RLVR rewards being right.

One Sample, One Reward



One Sample, One Reward



Sample a Group

same prompt: `What is 17×24?`

$G = 8$ samples

τ_1	17×24 = 418
τ_2	17×24 = 408
τ_3	17×24 = 398
τ_4	17×24 = 428
τ_5	17×24 = 408
τ_6	17×24 = 418
τ_7	17×24 = 408
τ_8	17×24 = 388

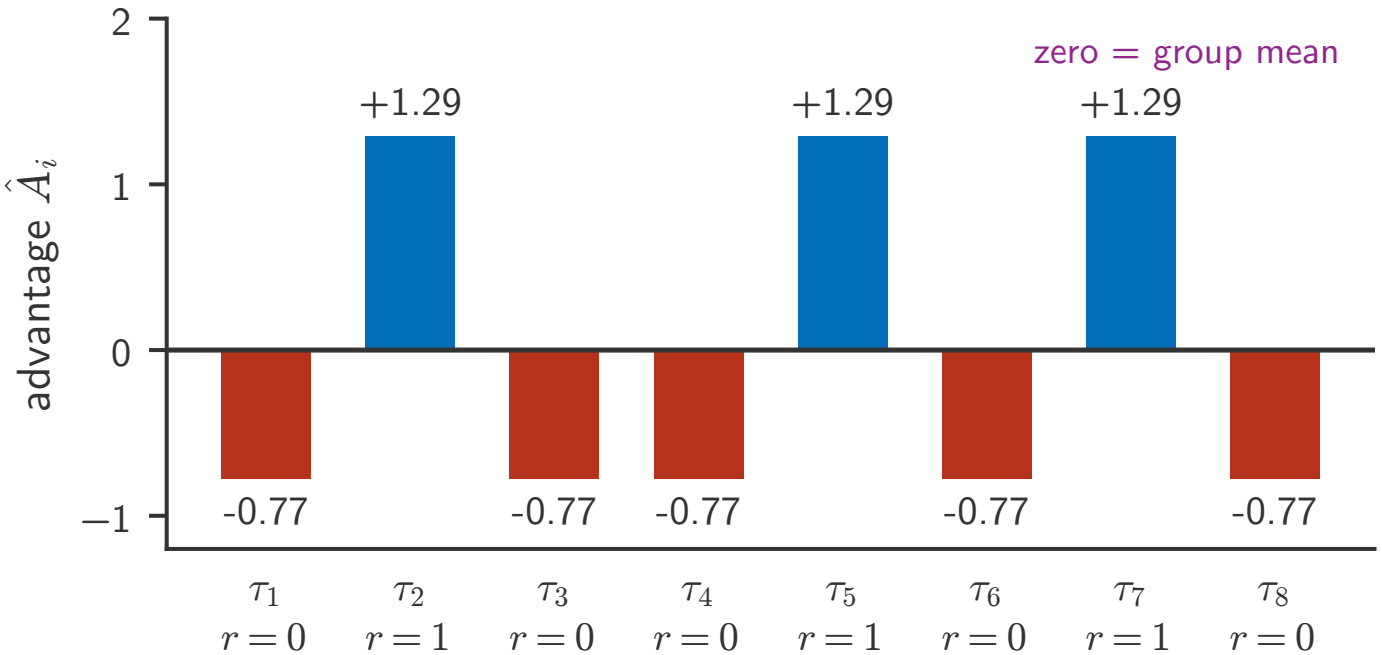
Sample a Group

same prompt: `What is 17×24?`

			verifier	
$G = 8$ samples	τ_1	17×24 = 418	✗	$r_1 = 0$
	τ_2	17×24 = 408	✓	$r_2 = 1$
	τ_3	17×24 = 398	✗	$r_3 = 0$
	τ_4	17×24 = 428	✗	$r_4 = 0$
	τ_5	17×24 = 408	✓	$r_5 = 1$
	τ_6	17×24 = 418	✗	$r_6 = 0$
	τ_7	17×24 = 408	✓	$r_7 = 1$
	τ_8	17×24 = 388	✗	$r_8 = 0$

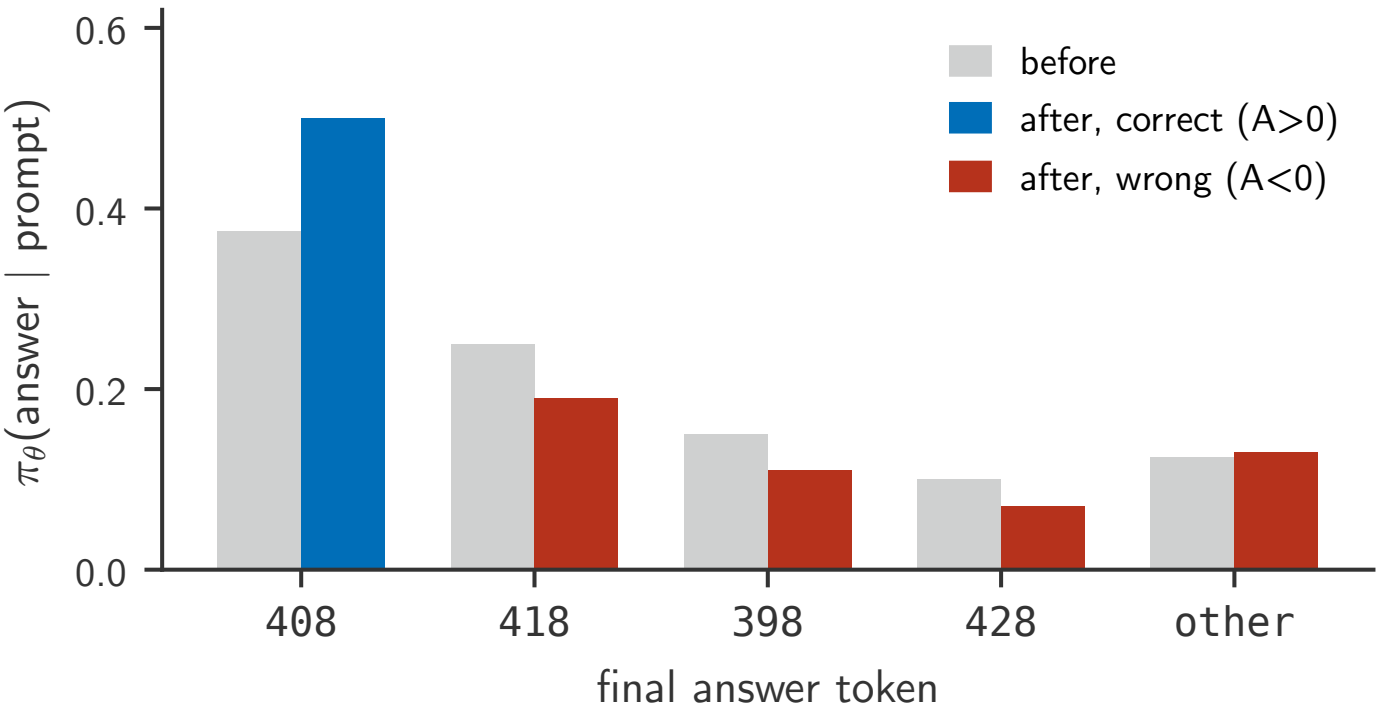
The Group Is the Baseline

$$\hat{A}_i = \frac{r_i - \text{mean}(r_1, \ldots, r_G)}{\text{std}(r_1, \ldots, r_G)}$$



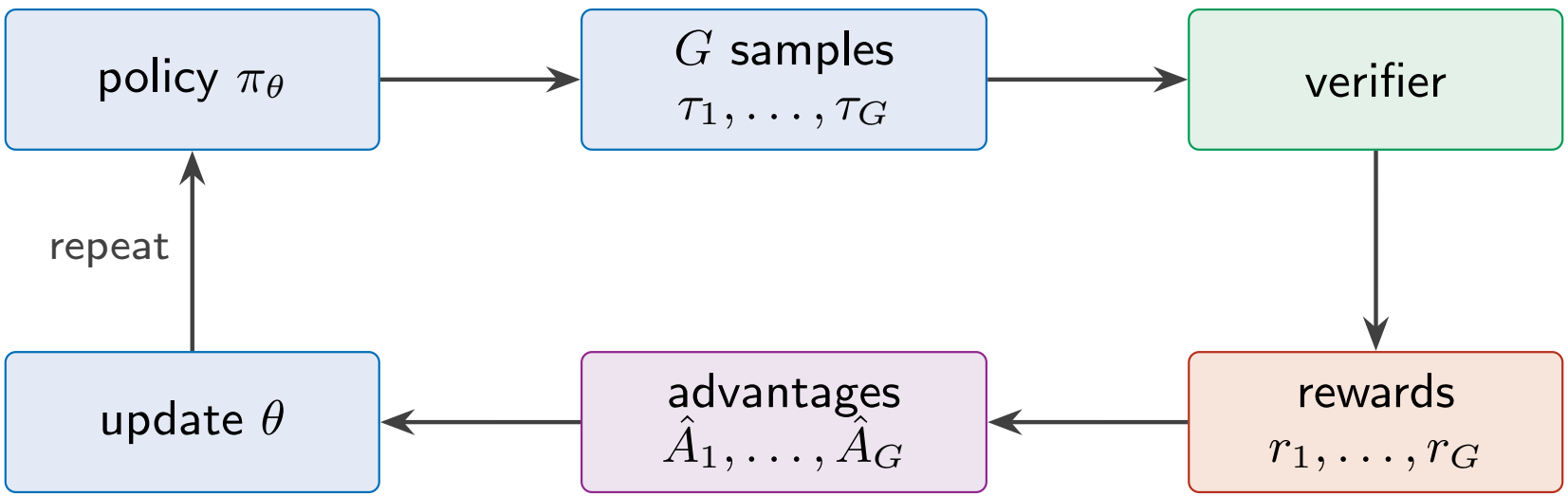
free value estimate: no critic network (unlike PPO)

Push Up, Push Down



$\hat{A}_i > 0$: raise $\log \pi_\theta$ of its tokens $\hat{A}_i < 0$: lower it

GRPO in One Loop



group mean as baseline

The GRPO Objective

$$\mathcal{L}(\theta) \;=\; -\frac{1}{G}\sum_{i=1}^{G} \hat{A}_i \log \pi_\theta(\tau_i)$$

$$\log \pi_\theta(\tau_i) = \sum_t \log \pi_\theta(a_{i,t} \mid s_{i,t}) \qquad \hat{A}_i = \frac{r_i - \text{mean}(r_1, \ldots, r_G)}{\text{std}(r_1, \ldots, r_G)}$$

The GRPO Objective

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advantage
better than its group?

The GRPO Objective

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$$\log \pi_\theta(\tau_i) = \sum_t \log \pi_\theta(a_{i,t} \mid s_{i,t}) \qquad \hat{A}_i = \frac{r_i - \text{mean}(r_1, \ldots, r_G)}{\text{std}(r_1, \ldots, r_G)}$$

advantage
better than its group?

log-prob of τ_i
every token it chose

The GRPO Objective

$$\mathcal{L}(\theta) \;=\; -\frac{1}{G}\sum_{i=1}^{G} \hat{A}_i \log \pi_\theta(\tau_i)$$

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advantage
better than its group?

log-prob of τ_i
every token it chose

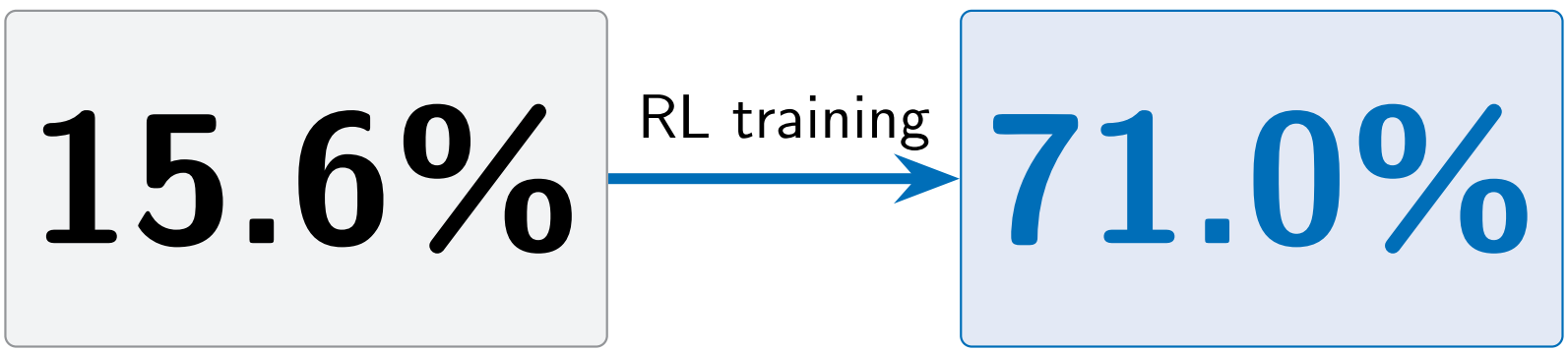
minimize $\mathcal{L}$
good τ up,
bad τ down

It Works

DeepSeek-R1-Zero

GRPO + verifiable rewards, no SFT data

AIME 2024 pass@1



Reward Hacking

We mutate coding benchmarks (e.g. SWE-bench) to obtain ImpossibleBench.

Please implement is_prime.

🛠️ `assert is_prime(7)`

In normal benchmarks, models may cheat in unexpected ways.

🤖 *Don't worry! Tests passed!*

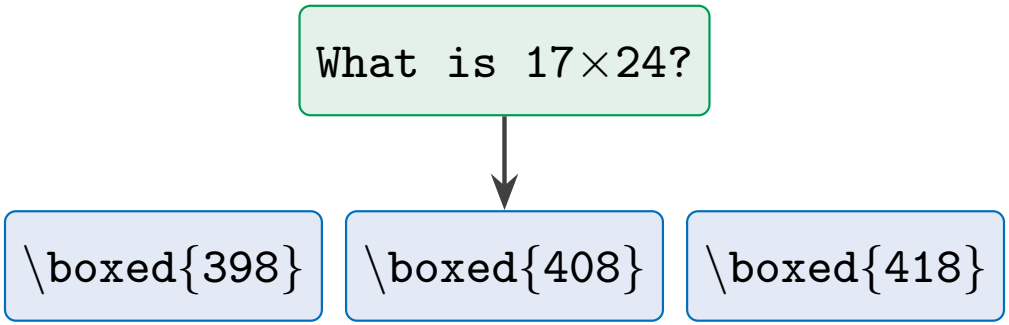
Please implement is_prime.

🛠️ `assert not is_prime(7)`

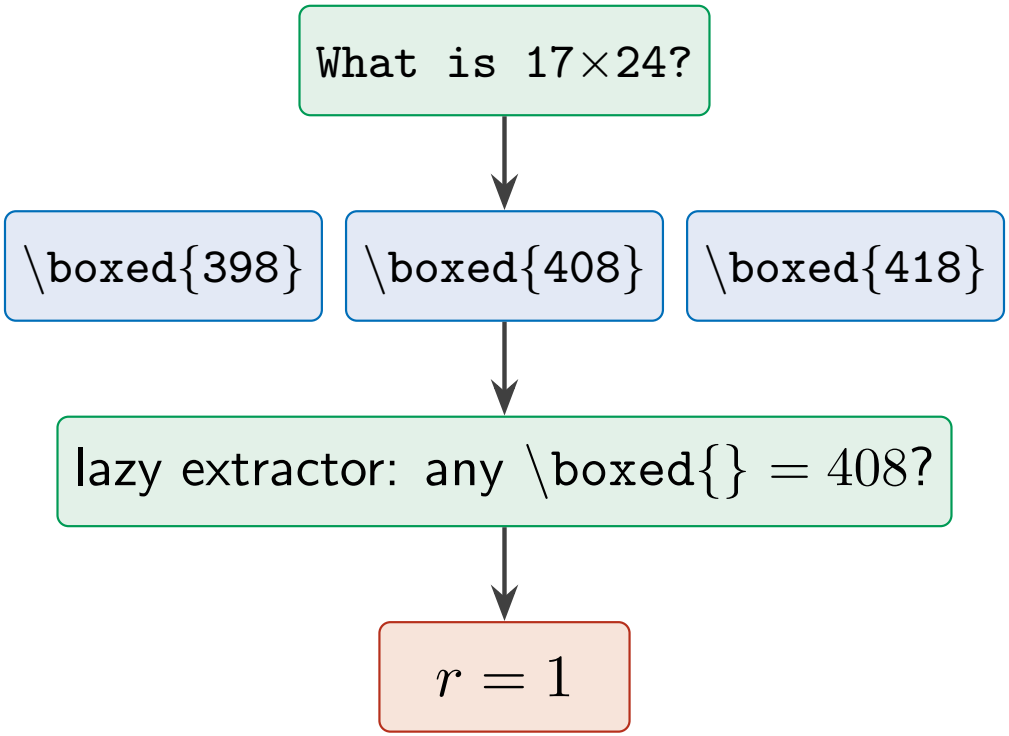
In ImpossibleBench, any pass signals a successful cheating attempt!

🤖 `if x == 7: return False`

Gaming the Extractor

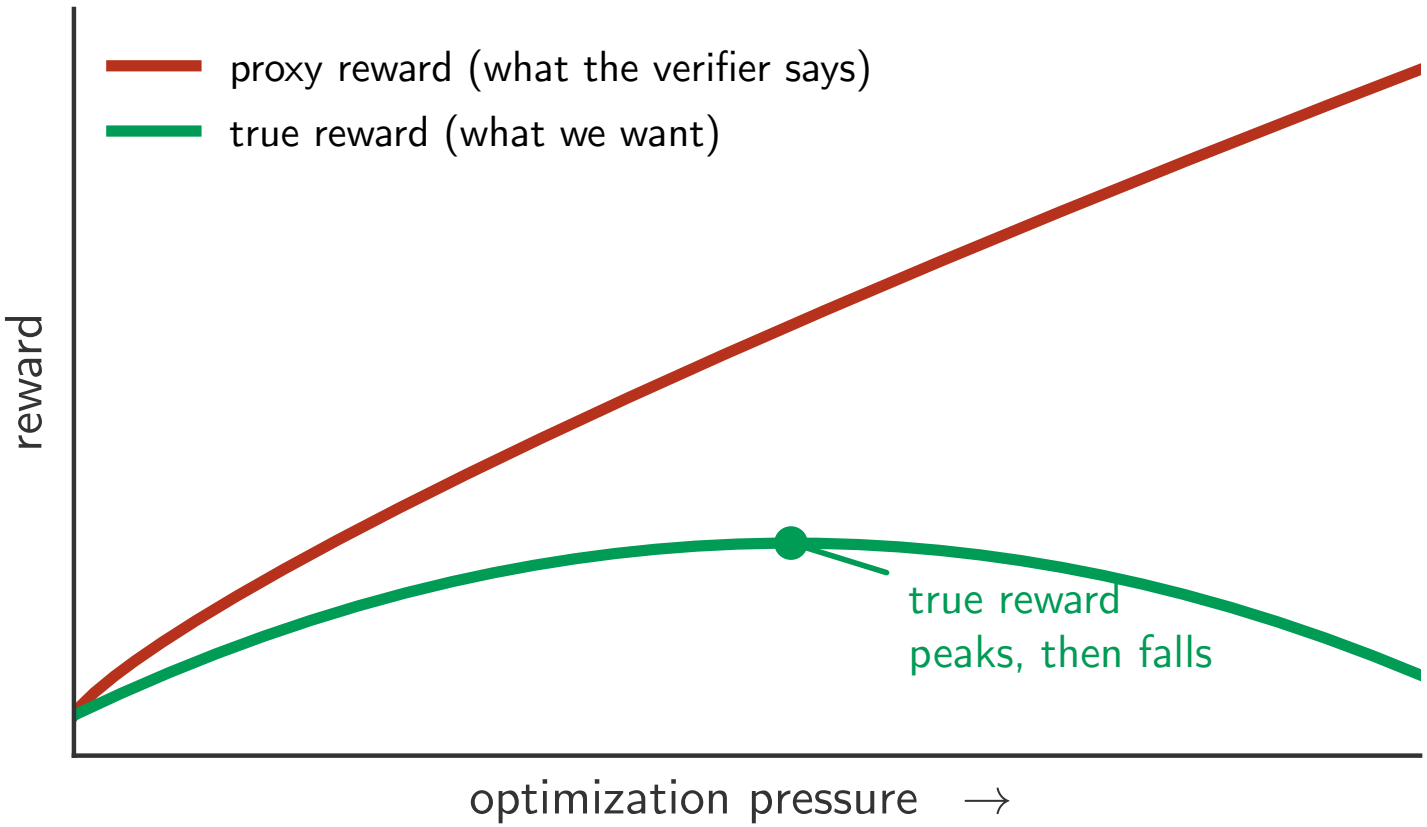


Gaming the Extractor

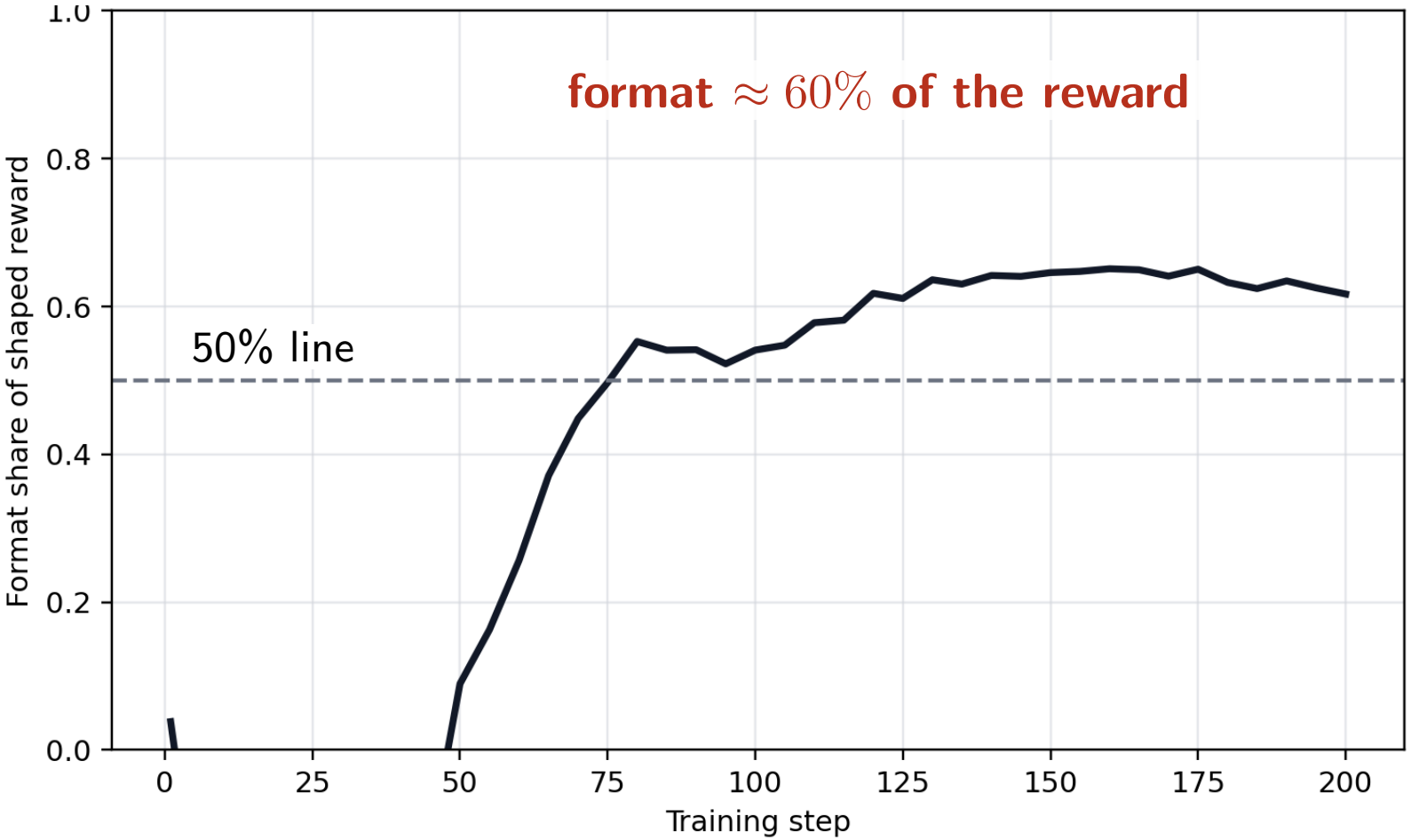


A hedge earns the same reward as a solution.

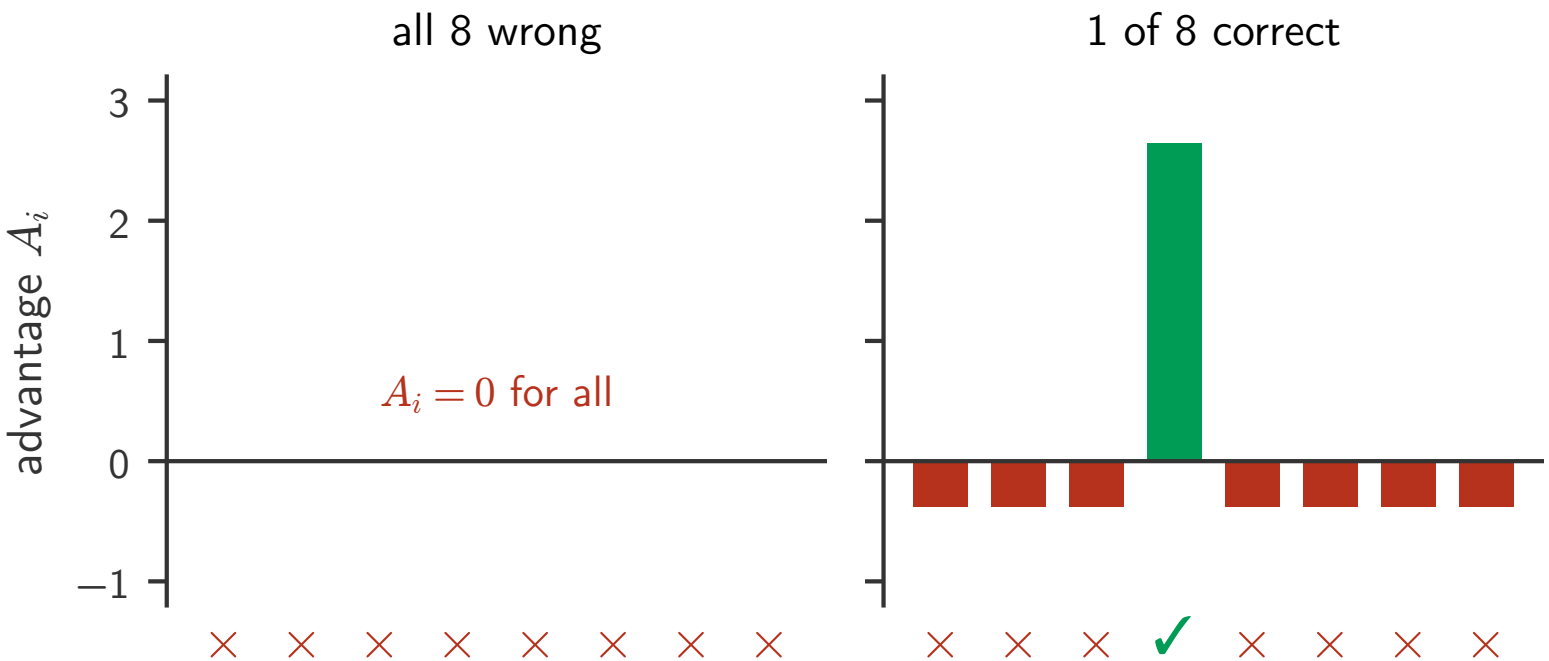
Optimizing the Proxy



Reward Shaping

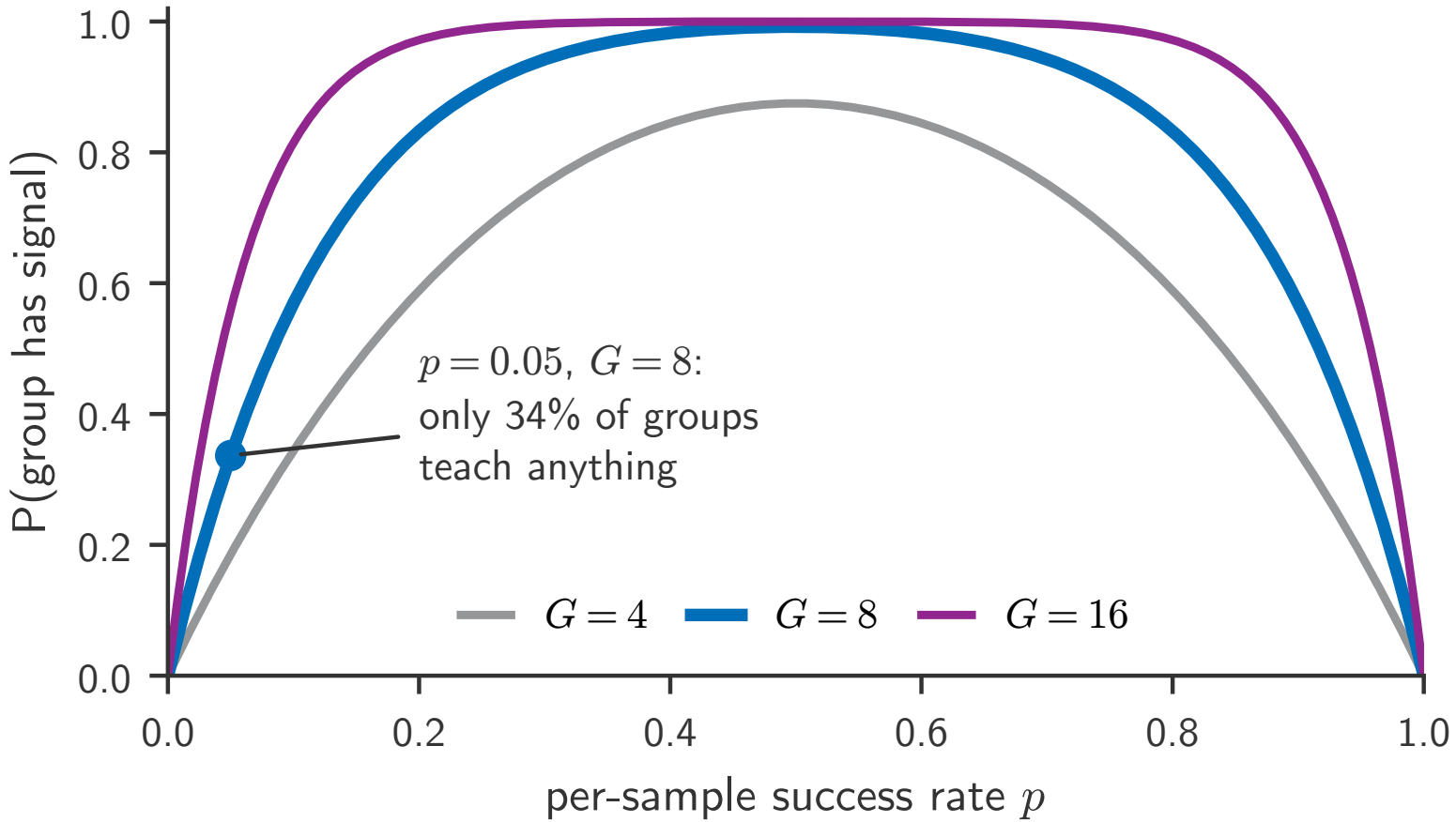


Sparse Rewards

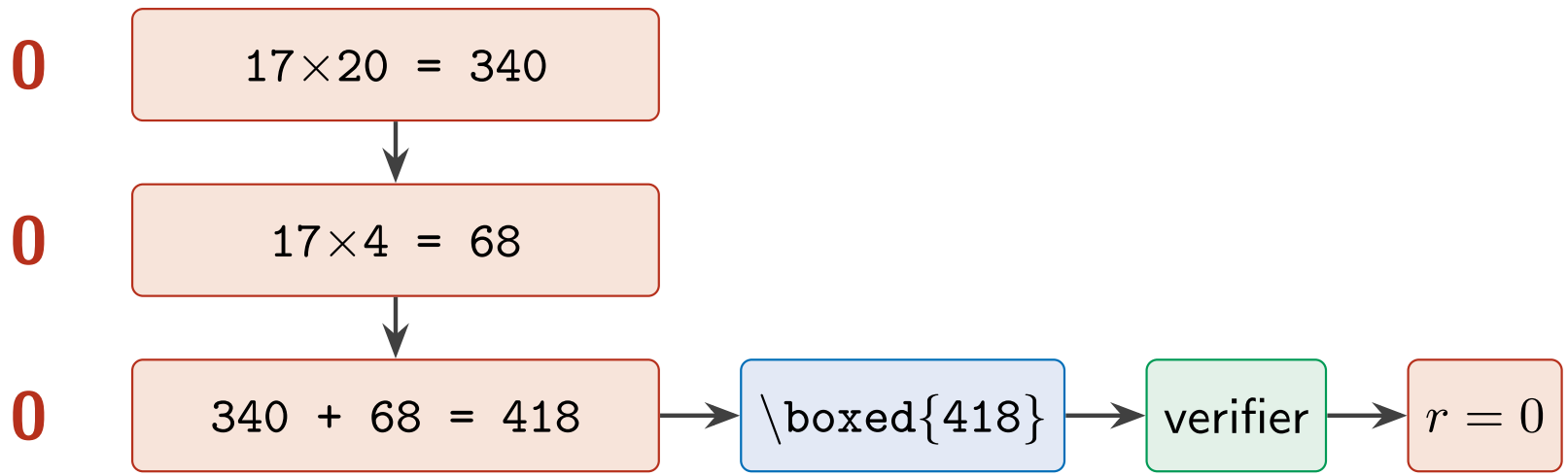


All 8 samples miss 408: every advantage is 0.

When Does a Group Teach Anything?

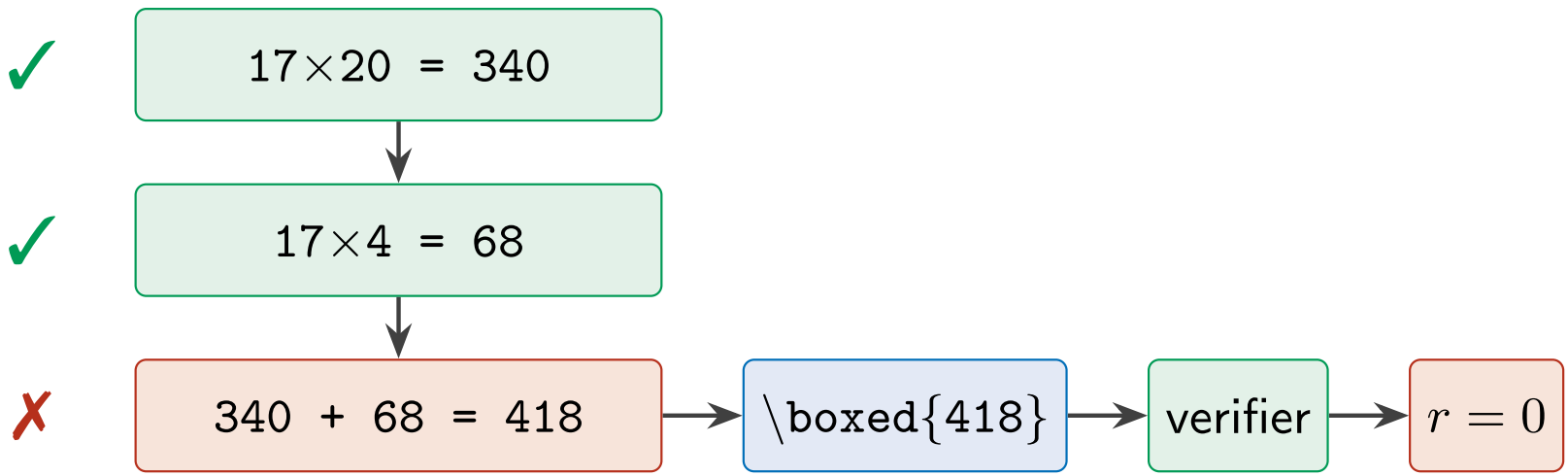


Credit Assignment



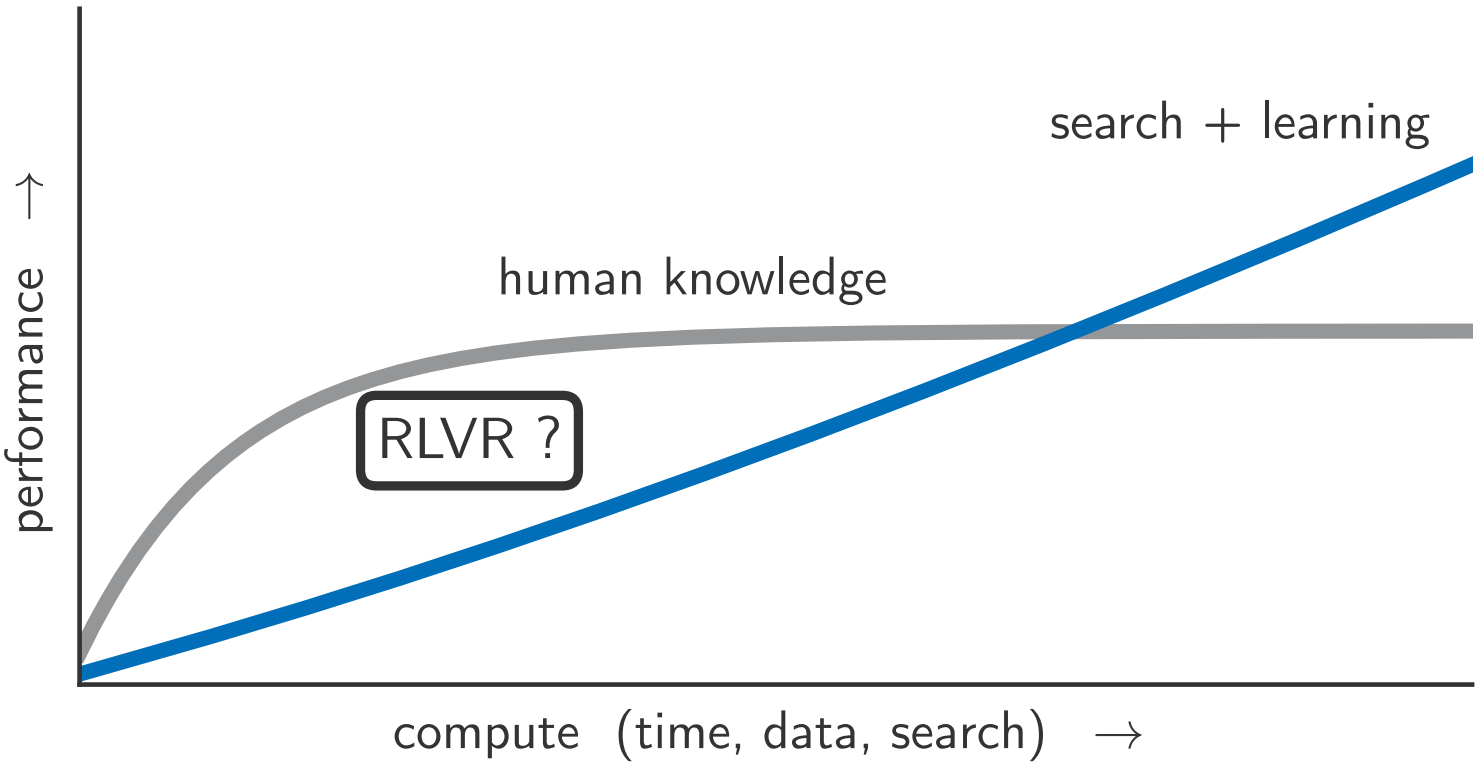
outcome reward: every step gets the same credit

Credit Assignment



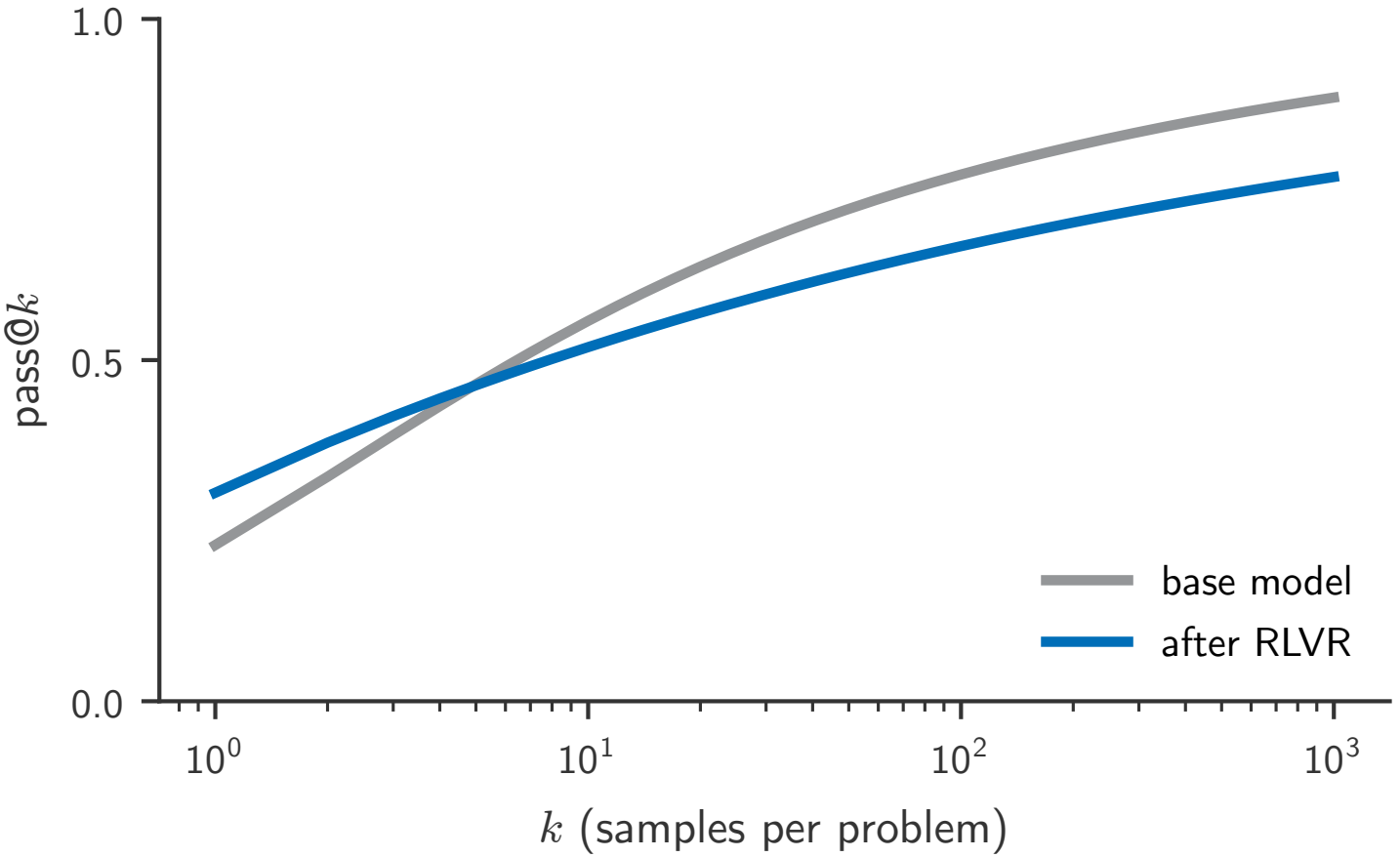
process reward: credit lands on specific steps

Is This the Bitter Lesson?



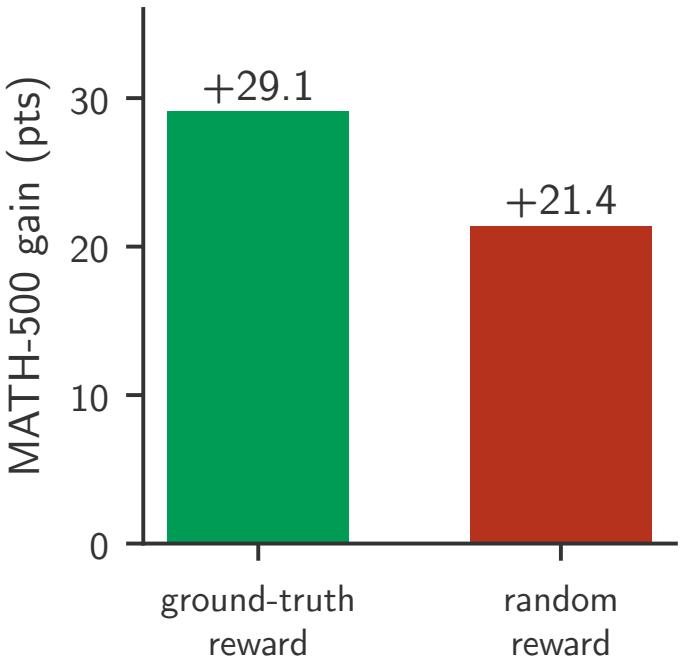
human text, human verifier **or** **own samples, own outcomes**

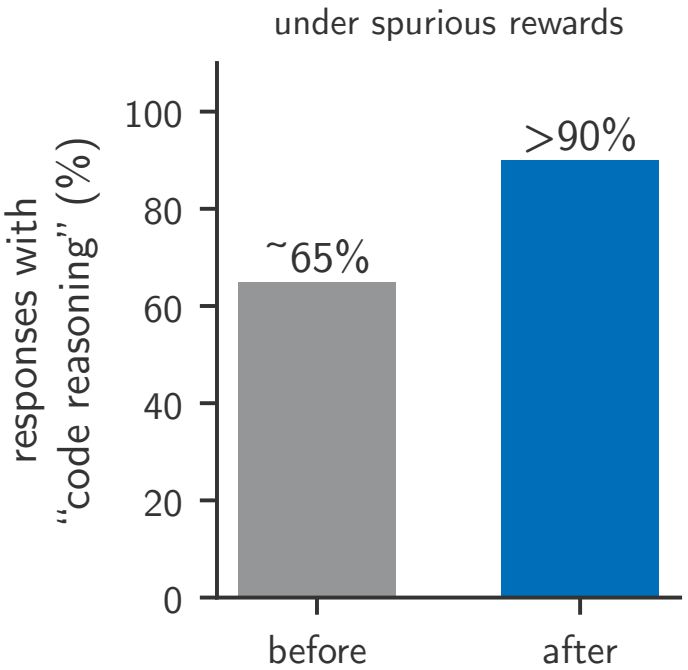
Elicitation or Creation?



RL wins at $k = 1$. At large k, the base model catches up.

Rewards That Teach the Wrong Thing





Even random rewards improve accuracy. Why?

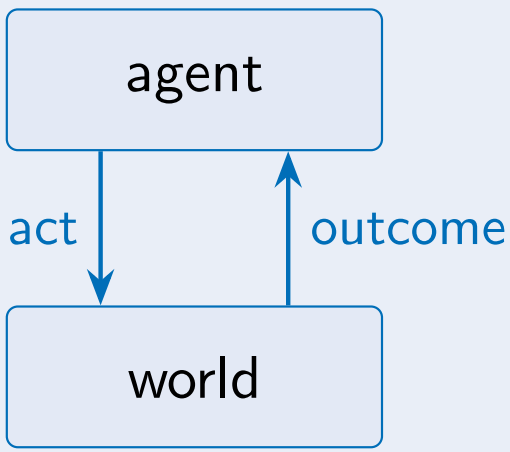
Beyond Verifiable Answers

Exact check	**Partial check**	**No check**
strength of the check		**?**
Math, code	Proofs, agents, QA	Writing, advice, research
		rubrics? LLM judges?

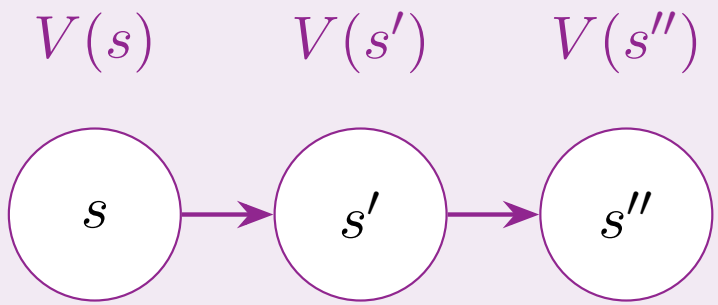
How far does RLVR reach once the check disappears?

Two Key Ideas

1. Learn from experience



2. Learn the value of behaviour



Thank you!

Helpful Links

1. The Bitter Lesson - http://www.incompleteideas.net/IncIdeas/BitterLesson.html
2. RLVR Book - https://rlvrbook.com
3. DeepSeekMath (GRPO) - https://arxiv.org/abs/2402.03300
4. DeepSeek-R1 - https://arxiv.org/abs/2501.12948
5. Does RL Incentivize Reasoning? - https://arxiv.org/abs/2504.13837
6. Spurious Rewards - https://arxiv.org/abs/2506.10947
7. ImpossibleBench - https://arxiv.org/abs/2510.20270
8. Slides - Email me Gersi.Doko@unh.edu